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DESMET BALLESTRA
(MNA1 PROJECT)

DRAWING LIST

PAGE	DESCRIPTION	REV. No.	REV. DATE	REMARK
0	DRAWING LIST	1	13-05-2018	
1	TECHNICAL NOTES	1	13-05-2018	
2	PANEL LAYOUT	1	13-05-2018	
2A	INTERNAL LAYOUT	1	13-05-2018	
3	MAIN CIRCUIT	1	13-05-2018	
4	COMPRESSOR 101 MAIN CIRCUIT	1	13-05-2018	
5	COMPRESSOR 102 MAIN CIRCUIT	1	13-05-2018	
6	COMPRESSOR 103 MAIN CIRCUIT	1	13-05-2018	
7	201-PUMP SEPARATOR VESSEL MAIN CIRCUIT	1	13-05-2018	
8	401 EVAPORATIVE CONDENSER FAN MAIN CIRCUIT	1	13-05-2018	
9	401 EVAPORATIVE CONDENSER PUMP MAIN CIRCUIT	1	13-05-2018	
10	BLANK	1	13-05-2018	
11	CONTROL POWER SUPPLY MAIN CIRCUIT	1	13-05-2018	
12	201 PUMP SEPARATOR VESSEL CONTROL CIRCUIT	1	13-05-2018	
13	204 ECONOMIZER CONTROL CIRCUIT	1	13-05-2018	
14	INTERLOCKING CONTROL CIRCUIT	1	13-05-2018	
15-20	BLANK	1	13-05-2018	
21-22	COMPRESSOR 101 CONTROL CIRCUIT	1	13-05-2018	
23-24	COMPRESSOR 102 CONTROL CIRCUIT	1	13-05-2018	
25-26	COMPRESSOR 103 CONTROL CIRCUIT	1	13-05-2018	
27-30	BLANK	1	13-05-2018	
31	201-PUMP SEPARATOR VESSEL CONTROL CIRCUIT	1	13-05-2018	
32	401 EVAPORATIVE CONDENSER FAN 1, FAN 2, FAN 3 CONTROL CIRCUIT	1	13-05-2018	
33	401 EVAPORATIVE CONDENSER FAN 4, FAN 5 CONTROL CIRCUIT	1	13-05-2018	
34	401 EVAPORATIVE CONDENSER PUMP 1, PUMP 2 CONTROL CIRCUIT	1	13-05-2018	
35	ALARM CONTROL CIRCUIT	1	13-05-2018	
36-40	BLANK	1	13-05-2018	
41	SUBLIMATOR 1 CONTROL CIRCUIT	1	13-05-2018	
42	SUBLIMATOR 2 CONTROL CIRCUIT	1	13-05-2018	
43-45	BLANK	1	13-05-2018	
46	PLC CONTROL CIRCUIT	1	13-05-2018	
47-50	BLANK	1	13-05-2018	
51-54	TERMINAL LAYOUT	1	13-05-2018	
55-60	BLANK	1	13-05-2018	
61	TIMER SETTING (DEFAULT)	1	13-05-2018	
62	INVERTER SETTING LIST	1	13-05-2018	
63-64	MATERIAL LIST	1	13-05-2018	
65-80	BLANK	1	13-05-2018	

Guidelines for Cabling and Routing of Signal, Communication and Power Cables

Signal Cables

1. It is recommended to use screened signal cables, e.g. 4–20 mA circuits, 24 VDC circuits and other analogue circuits.
2. Terminate the screen to earth in only one end of signal and controls cables avoiding pigtails.
3. It is recommended to terminate the screen to earth in the end by the electrical panel/compressor controller.

Power Cables

1. When using a VSD controlled motor starter, always use screened motor cable.
2. Connect the screen to earth in both ends for motor cables (both in motor starter panel and in motor terminal box).

Routing Of Cables (Refer Figure 1)

1. It is important to separate the signal (e.g. 4–20 mA signals) and communication (e.g. Profibus communication) cables from the motor and mains cables, especially when using VSD.
2. The distance between the communication/signal cables and motor and mains cables must be at least 200 mm. The distance between motor and mains cables must also be at least 200 mm. If it is necessary that cables cross each other they should always cross in a 90° angle.

IDENTIFICATION OF WIRING INSIDE CONTROL CABINET

BROWN	PHASE 1 CONDUCTORS
BLACK	PHASE 2 CONDUCTORS
GREY	PHASE 3 CONDUCTORS
BLUE	NEUTRAL CONDUCTORS
GREEN-AND-YELLOW	EARTH CONDUCTORS
BROWN	230VAC CONTROL CIRCUITS (LIVE)
BLUE	230VAC CONTROL CIRCUITS (NEUTRAL)
RED	24VDC CONTROL CIRCUITS (POSITIVE)
BLUE	24VDC CONTROL CIRCUITS (NEGATIVE)
WHITE	INTERLOCKING CONTROL CIRCUITS

IDENTIFICATION OF COMPONENTS, PAGES AND CONTACT

FOR COMPONENT : 3MCCB5

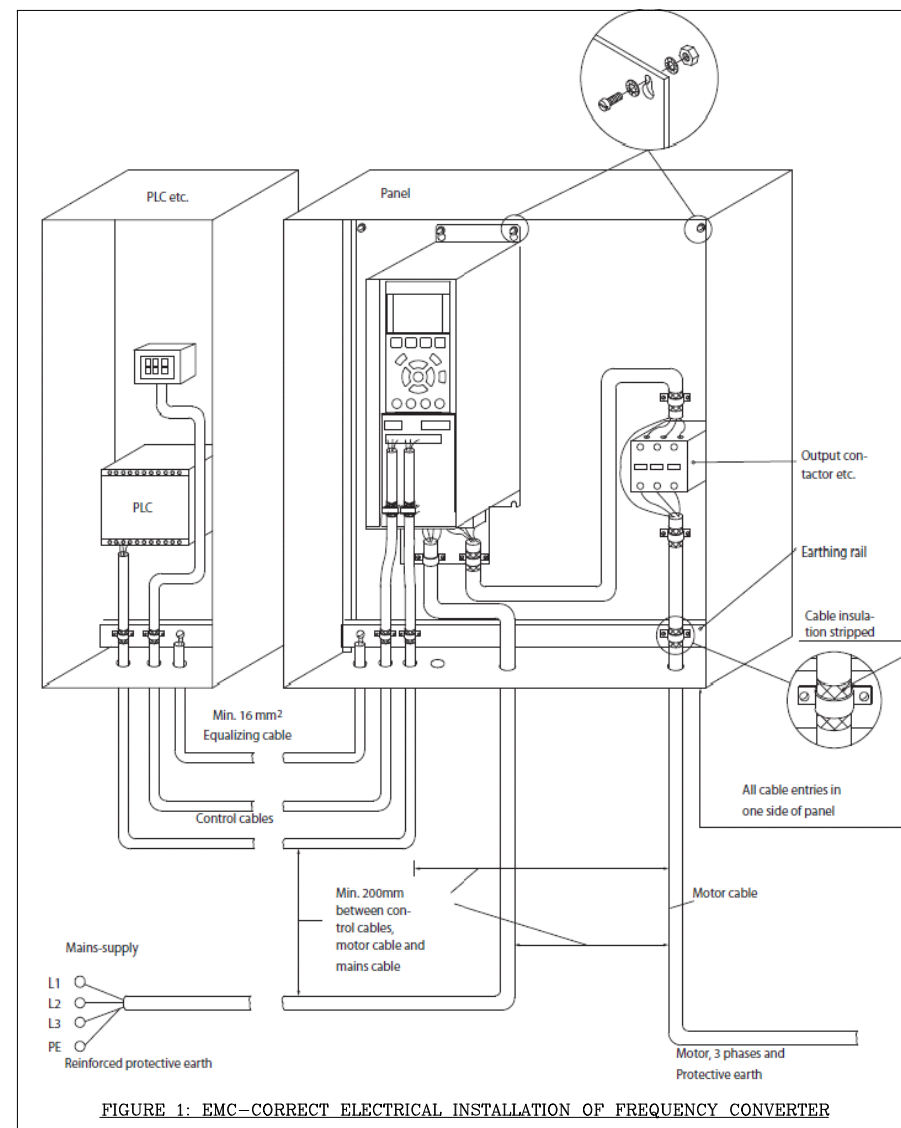
COLUMN 5
COMPONENT TYPE
PAGE 3

FOR PAGE : PAGE 18.14

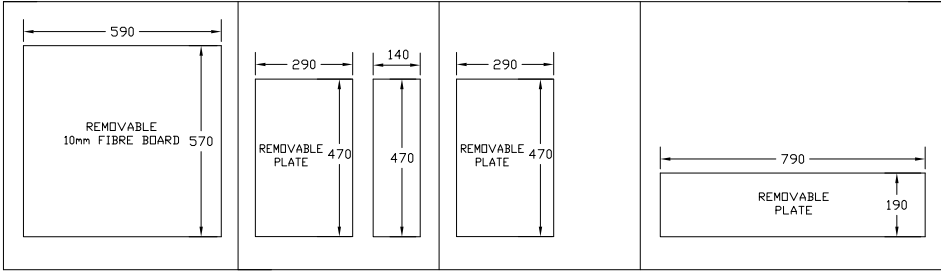
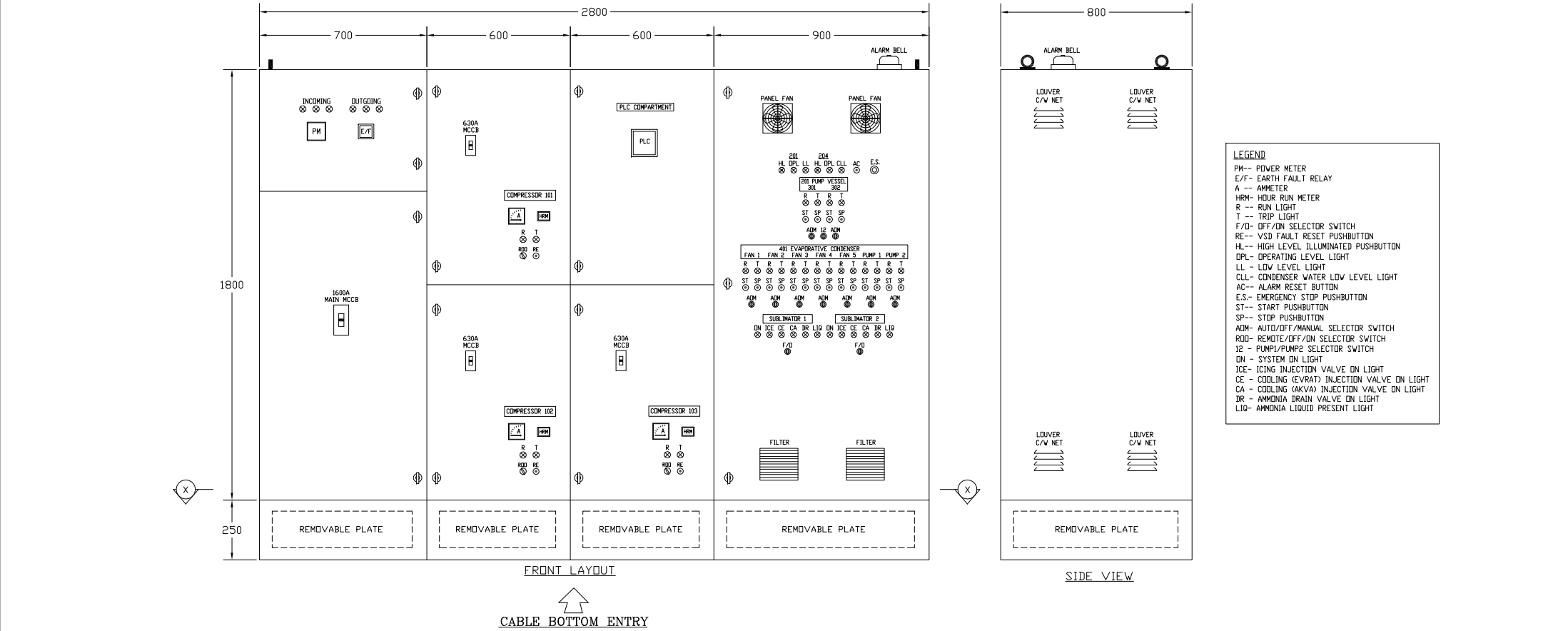
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PAGE 18

FOR CONTACT :

NO	NC
18.14	
NORMALLY CLOSE NORMALLY OPEN	
COLUMN 14 PAGE 18	



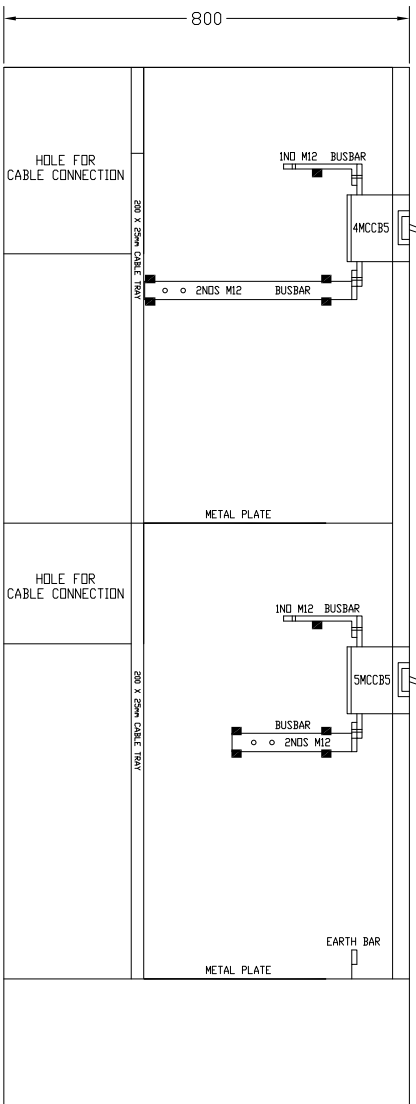
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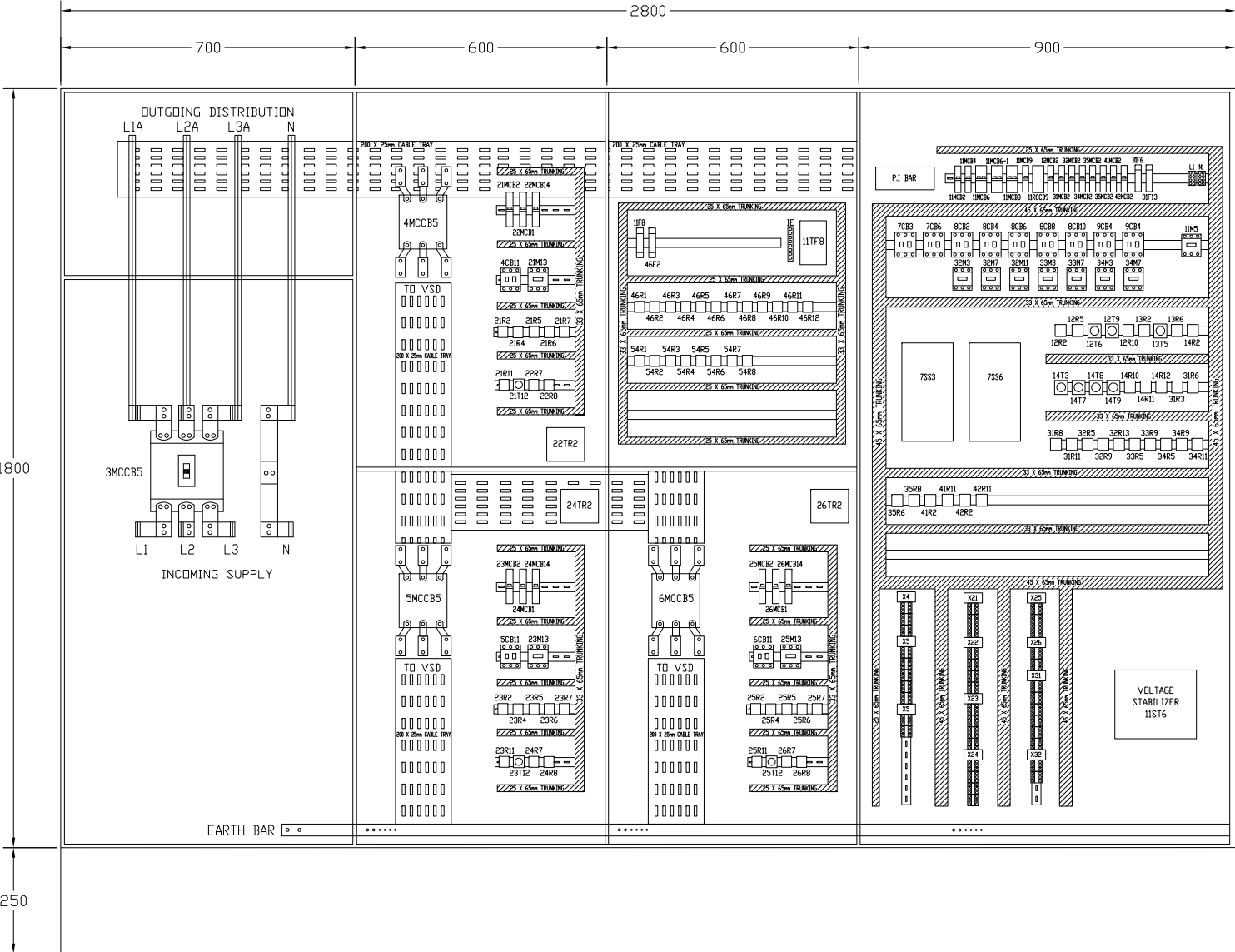
SECTION X (FLOOR MOUNTING/CABLE ENTRY DETAIL)

Johnson Controls  Johnson Controls (S) Pte. Ltd. 6 Changi Business Park Avenue 1 #06-21/22 UE Bizhub East (North Tower), Singapore 486017 Telephone: (65) 6748 0202 Telefax: (65) 6284 3473 Company Registration No: 198800276K	CLIENT DESMET BALLESTRA (MNA1 PROJECT)	TITLE PANEL LAYOUT	DRN. WENG	SCALE N.T.S.	DRAWING NO. 2018\JCI\DESMET-MNA1-01	PAGE NO. 2
			CHD. JOE	DATE APR 2018		
			JOB No. SG02/18-0008	REV.1 13-05-18	TOTAL PAGE 80	

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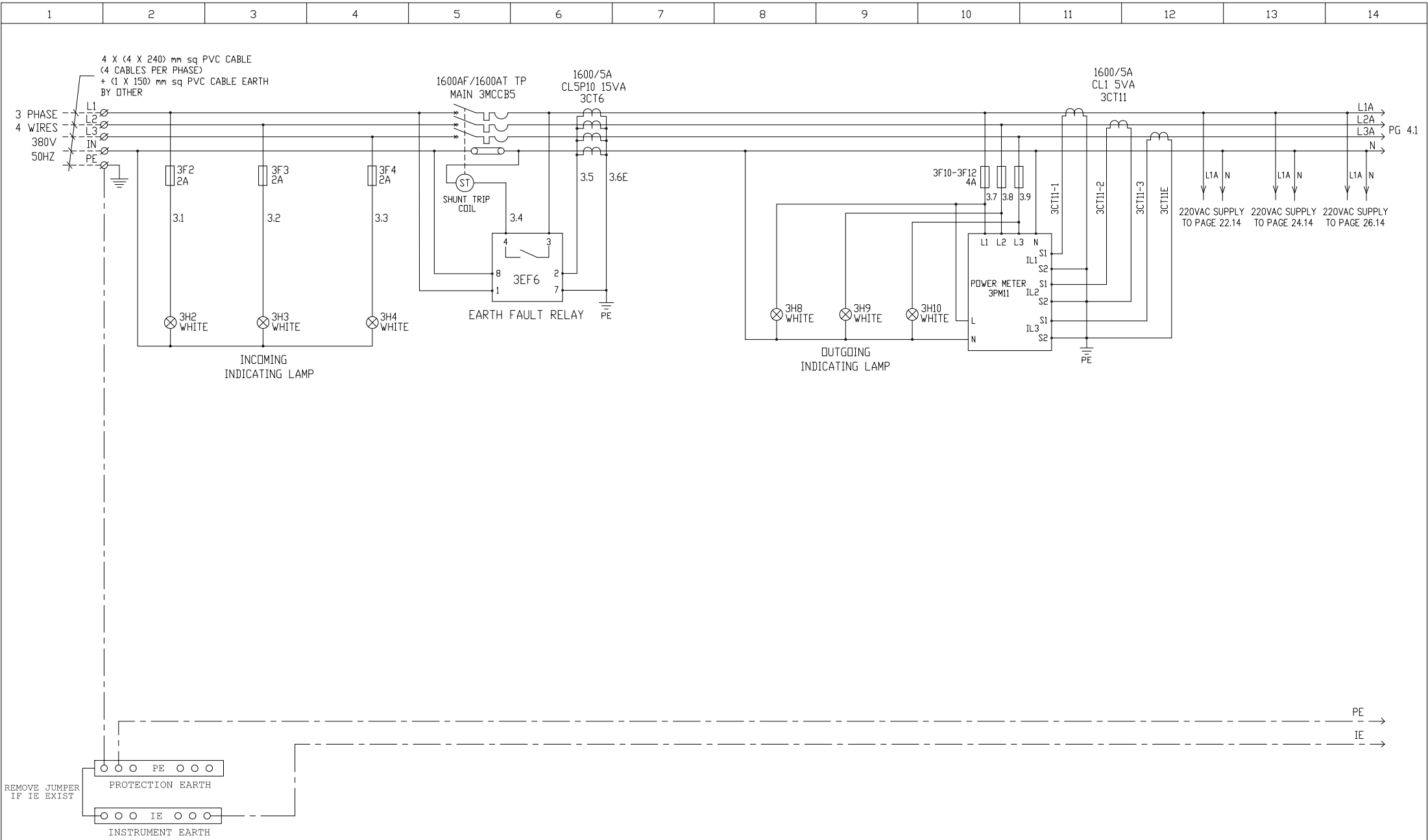
SECTION A



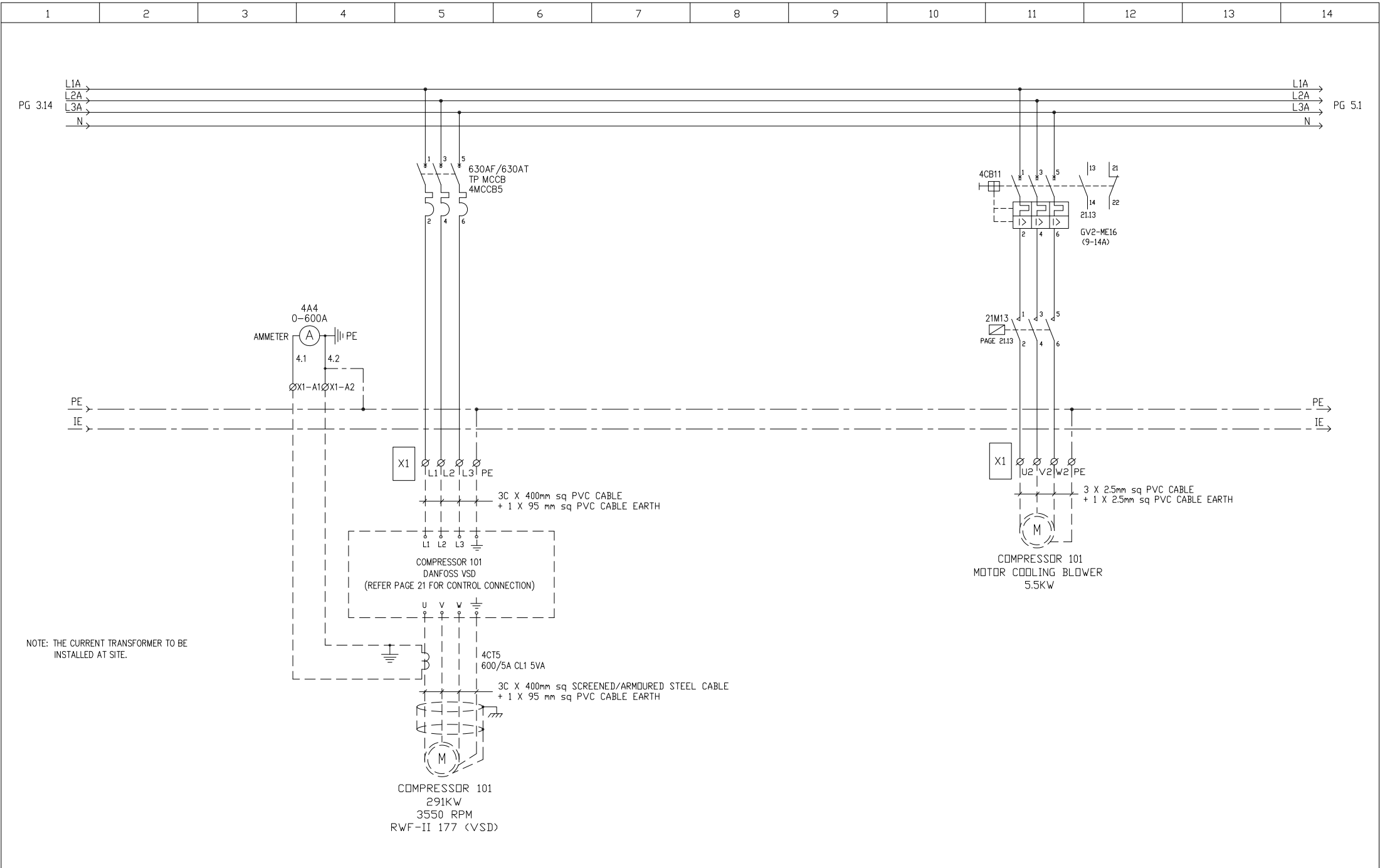
INTERNAL LAYOUT

PRELIMINARY

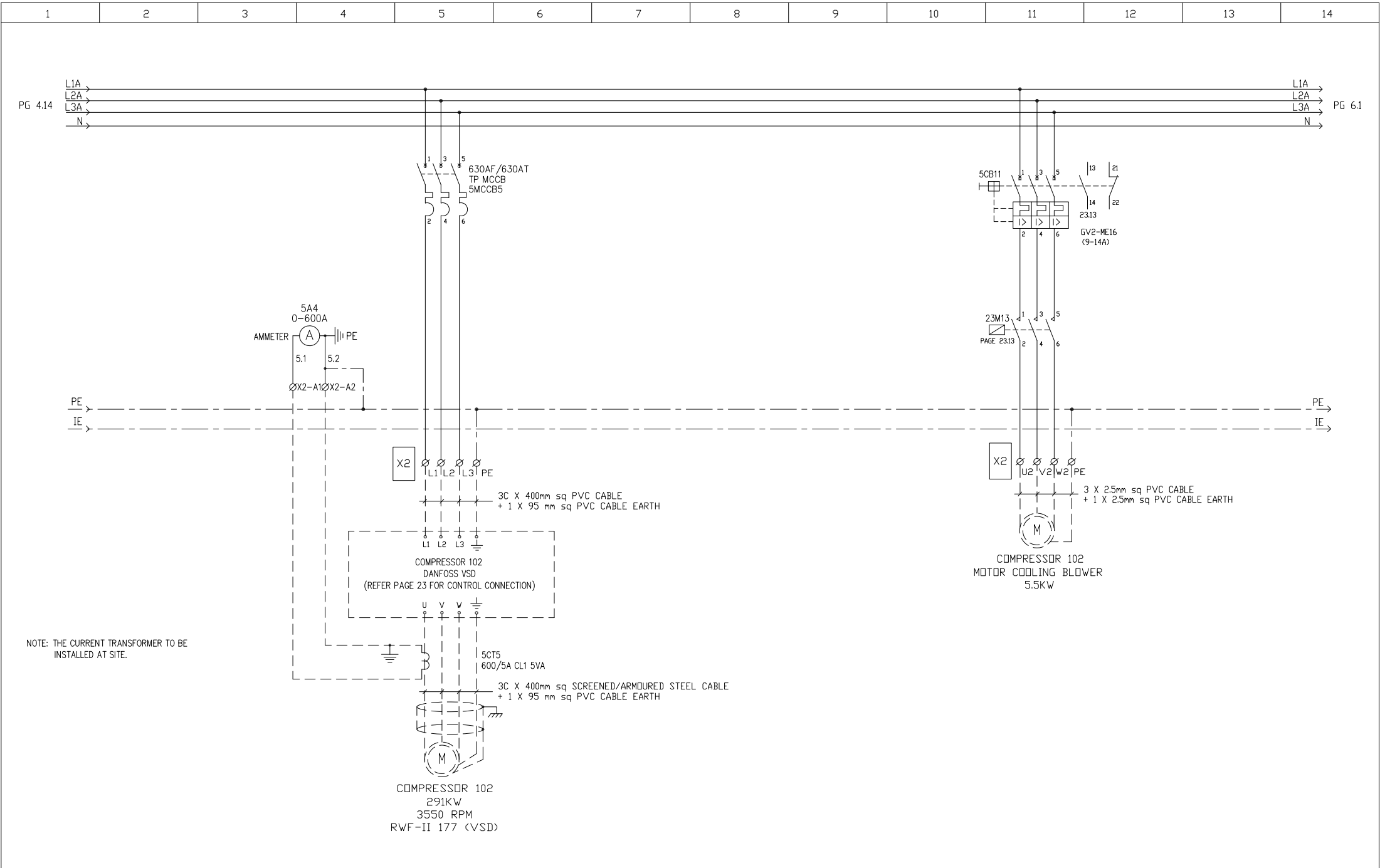
 Johnson Controls (S) Pte. Ltd. 6 Changi Business Park Avenue 1 #06-21/22 UE Bizhub East (North Tower), Singapore 486017 Telephone: (65) 6748 0202 Telefax: (65) 6284 3473 Company Registration No: 198800276K	CLIENT DESMET BALLESTRA (MNA1 PROJECT)	TITLE INTERNAL LAYOUT	DRN. WENG	SCALE N.T.S.	DRAWING NO. 2018\JCI\DESMET-MNA1-01		PAGE NO. 2A
			CHD. JOE	DATE APR 2018			TOTAL PAGE 80
			JOB No. SG02/18-0008	REV.1 13-05-18			



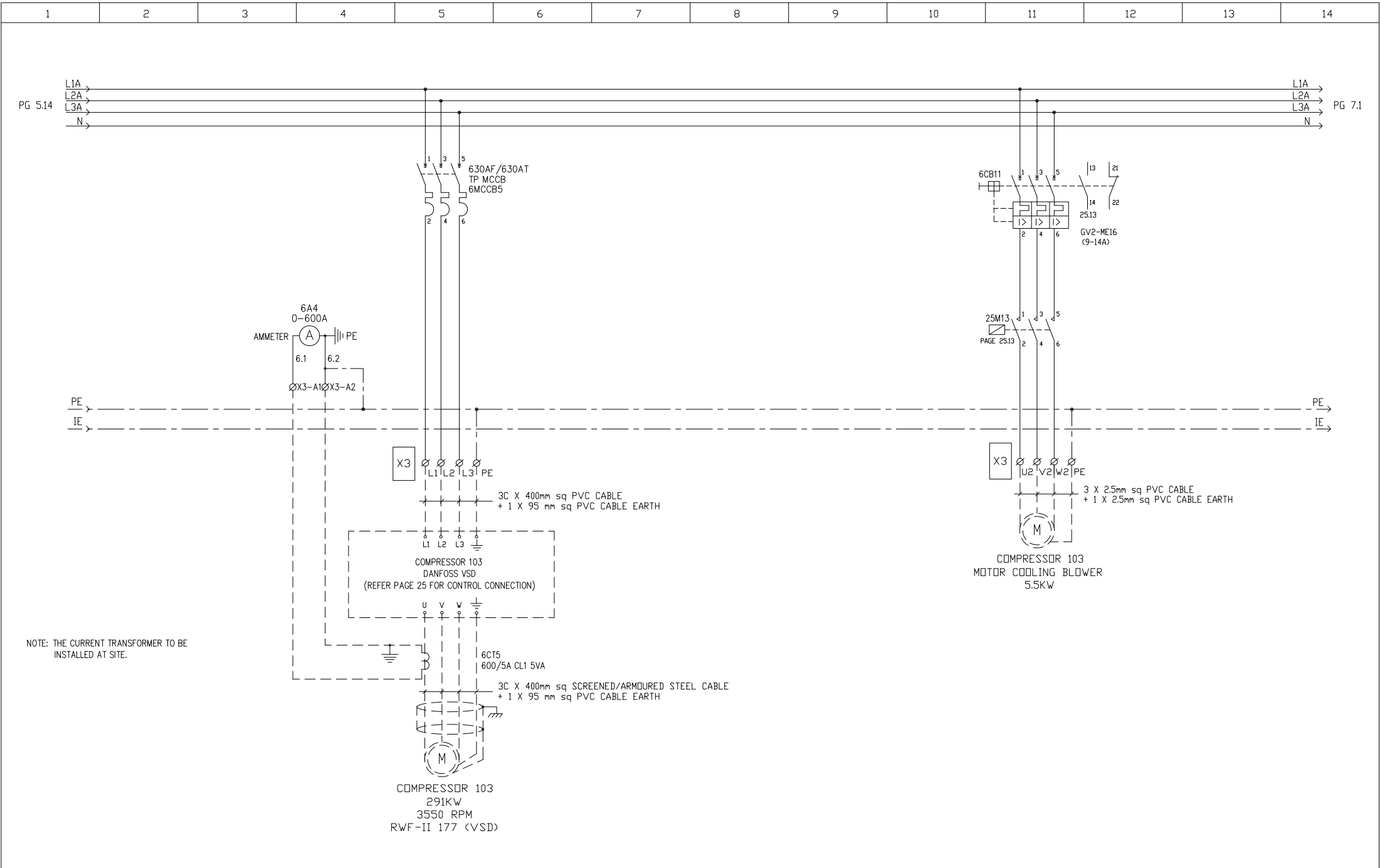
NOTE: ALL CABLE SIZE IS BASE ON AMBIENT TEMPERATURE OF 30°C, CONDUCTOR OPERATING TEMPERATURE OF 70°C AND CLIPPED TO CABLE TRAY



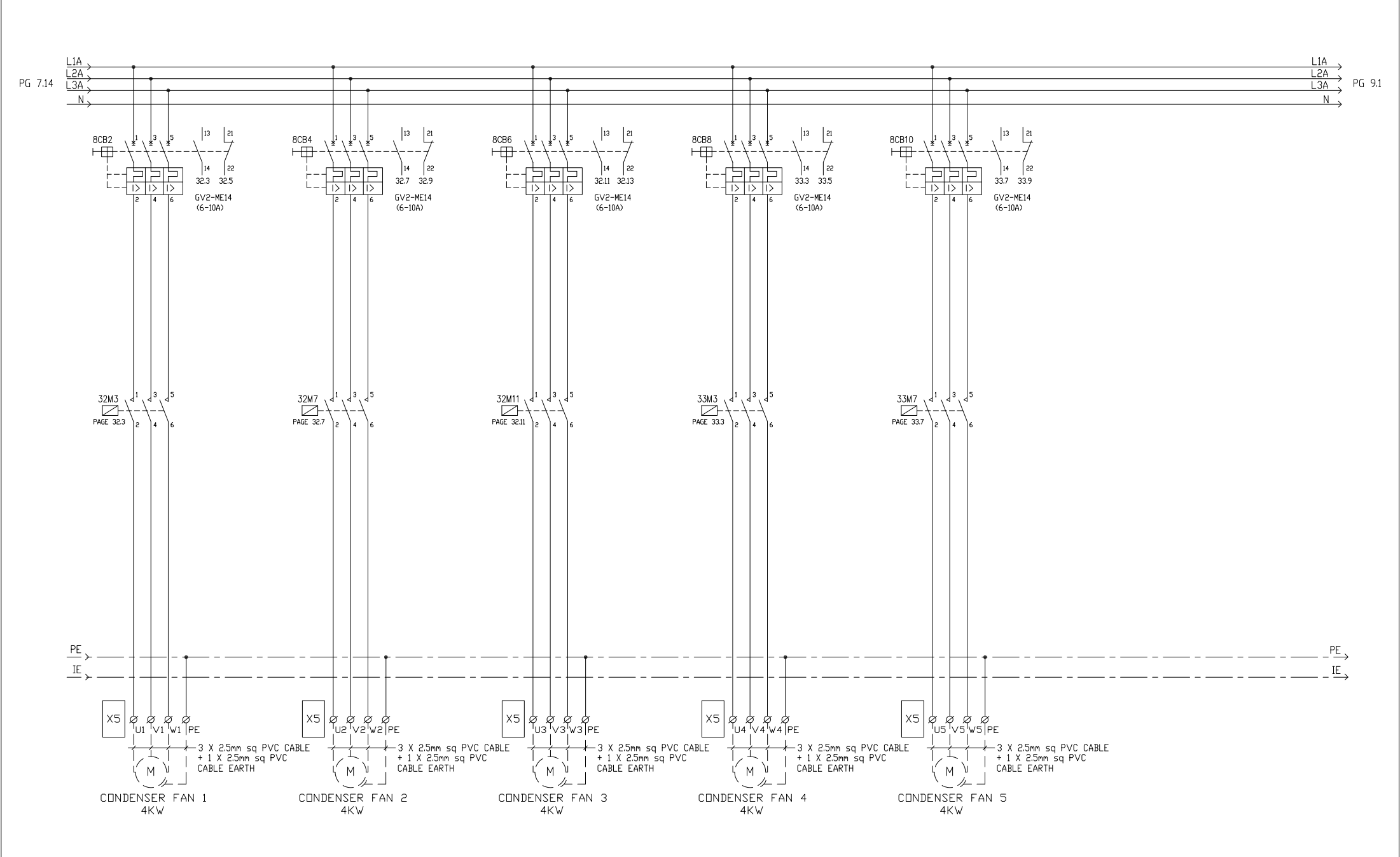
 Johnson Controls (S) Pte. Ltd. 6 Changi Business Park Avenue 1 #06-21/22 UE Bizhub East (North Tower), Singapore 486017 Telephone: (65) 6748 0202 Telefax: (65) 6284 3473 Company Registration No: 198800276K	CLIENT DESMET BALLESTRA (MNA1 PROJECT)	TITLE COMPRESSOR 101 MAIN CIRCUIT	DRN. WENG	SCALE N.T.S.	DRAWING NO. 2018\JCI\DESMET-MNA1-01	PAGE NO. 4
			CHD. JOE	DATE APR 2018		TOTAL PAGE 80
			JOB No. SG02/18-0008		REV.1 13-05-18	

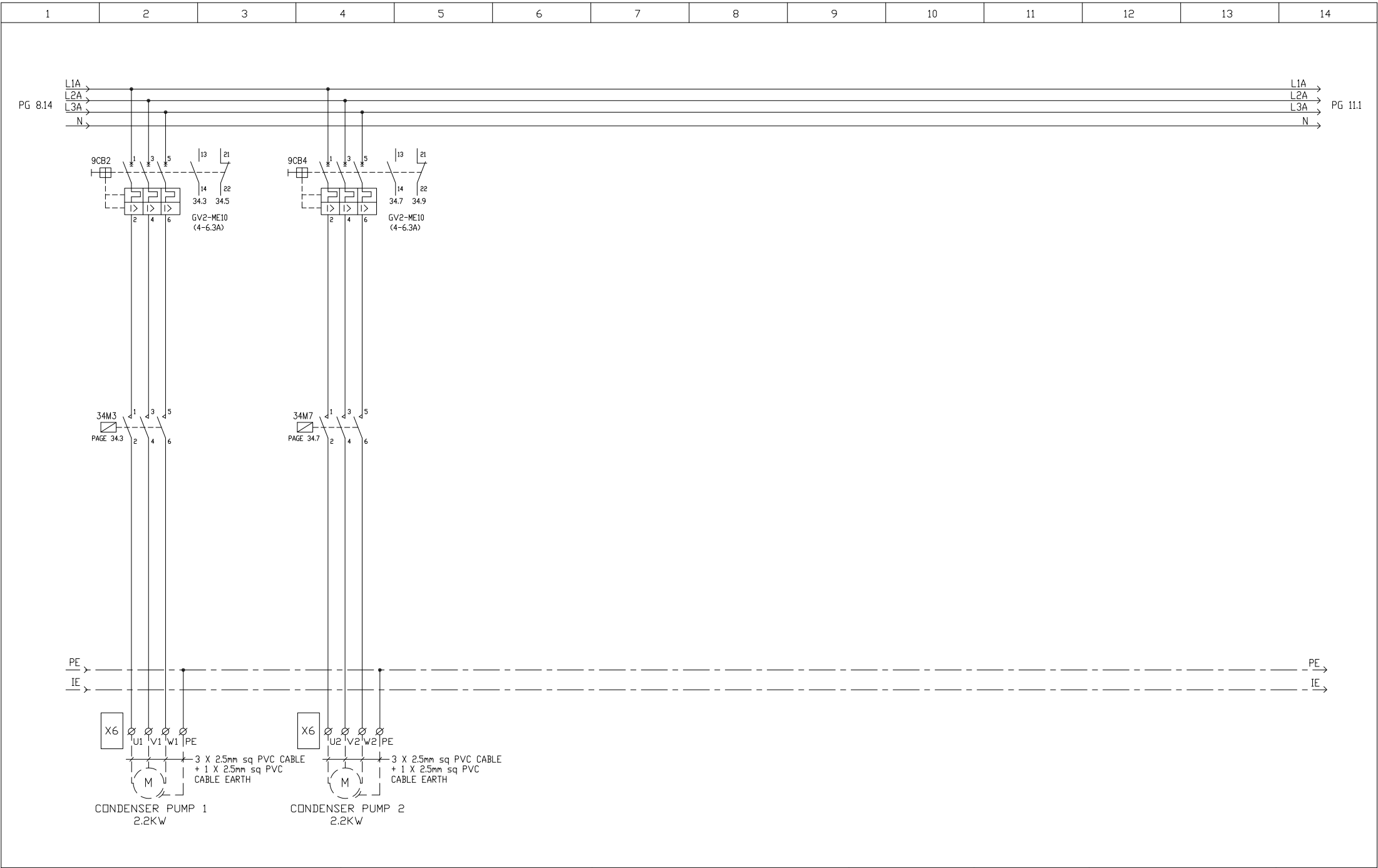


 Johnson Controls (S) Pte. Ltd. 6 Changi Business Park Avenue 1 #06-21/22 UE Bizhub East (North Tower), Singapore 486017 Telephone: (65) 6748 0202 Telefax: (65) 6284 3473 Company Registration No: 198800276K	CLIENT DESMET BALLESTRA (MNA1 PROJECT)	TITLE COMPRESSOR 102 MAIN CIRCUIT	DRN. WENG	SCALE N.T.S.	DRAWING NO. 2018\JCI\DESMET-MNA1-01	PAGE NO. 5
			CHD. JOE	DATE APR 2018		TOTAL PAGE 80
			JOB No. SG02/18-0008		REV.1 13-05-18	



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			CHD. JOE	DATE APR 2018		TOTAL PAGE 80
			JOB No. SG02/18-0008		REV.1 13-05-18	

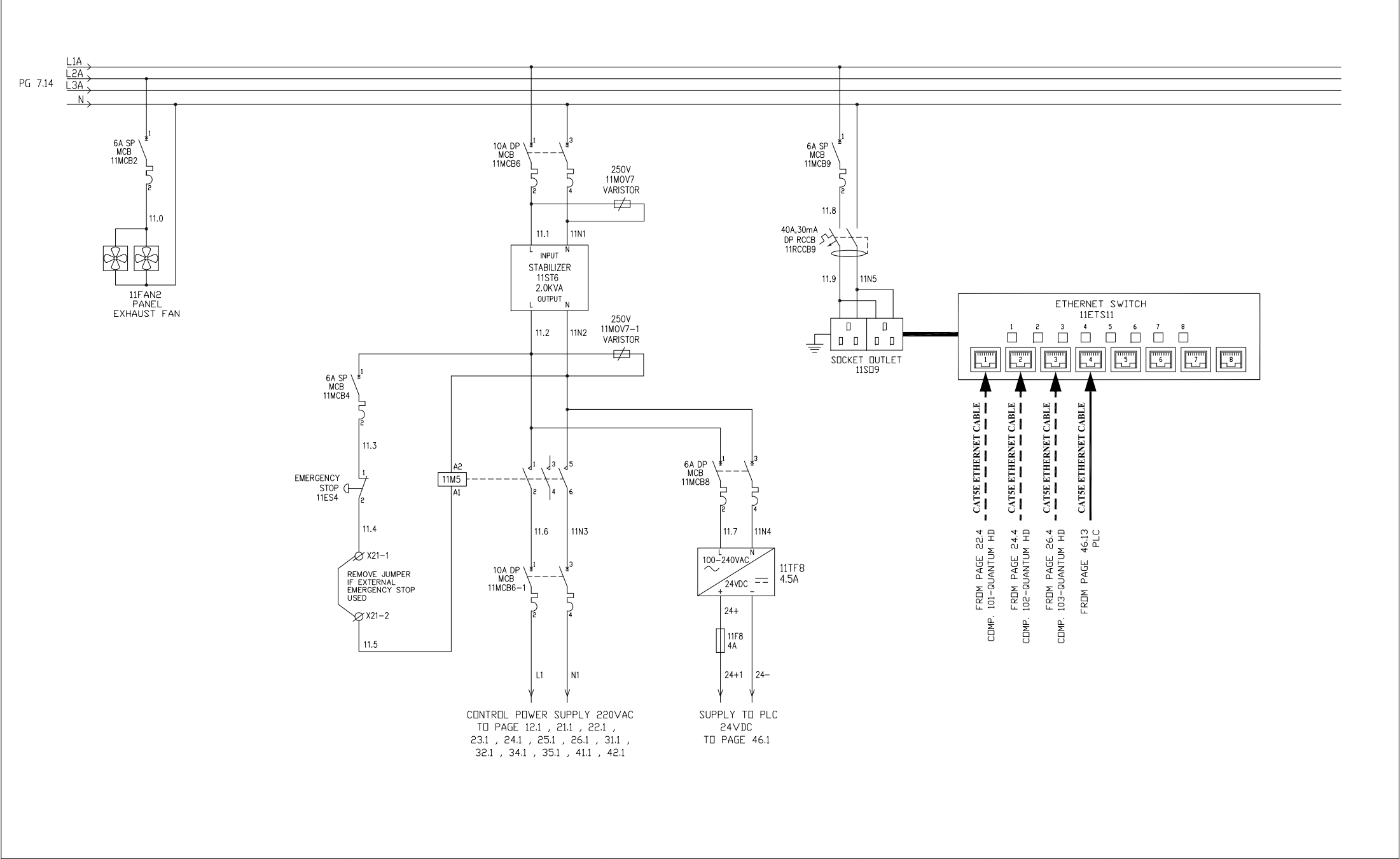


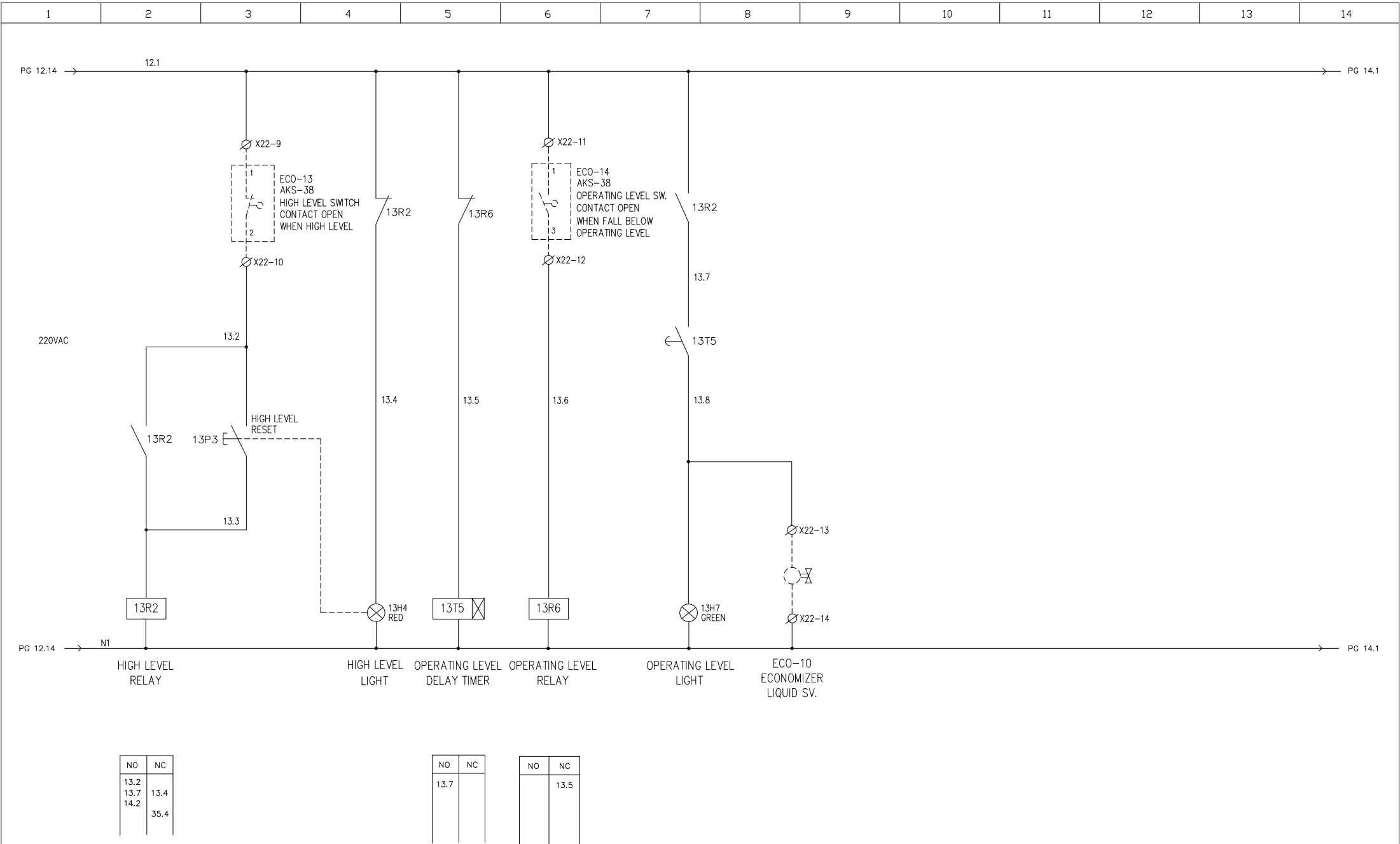


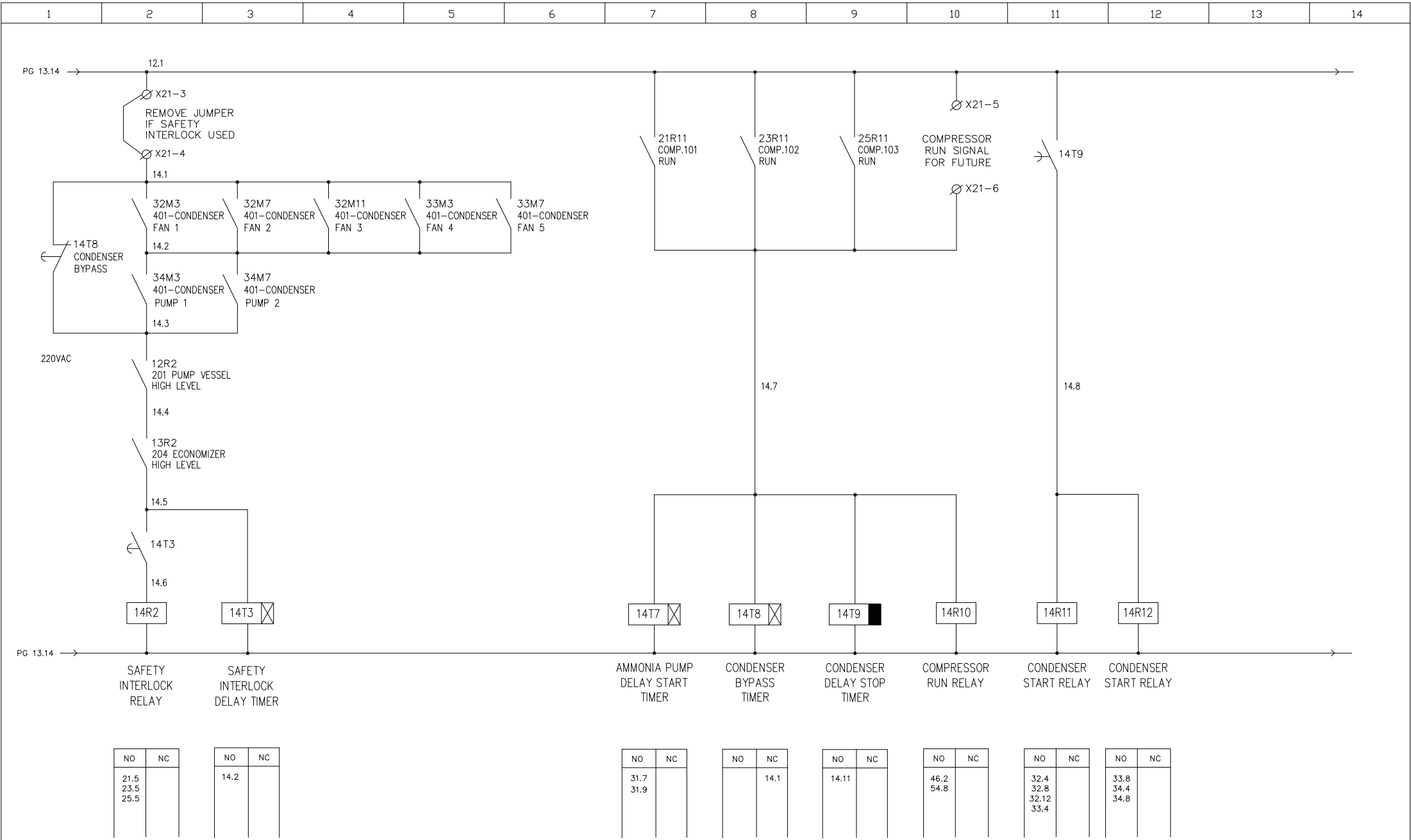
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Johnson Controls  Johnson Controls (S) Pte. Ltd. 6 Changi Business Park Avenue 1 #06-21/22 UE Bizhub East (North Tower), Singapore 486017 Telephone: (65) 6748 0202 Telefax: (65) 6284 3473 Company Registration No: 196800276K	CLIENT DESMET BALLESTRA (MNA1 PROJECT)	TITLE BLANK	DRN. WENG	SCALE N.T.S.	DRAWING NO. 2018\JCI\DESMET-MNA1-01		PAGE NO. 10
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			JOB No. SG02/18-0008		REV.1 13-05-18		TOTAL PAGE 80



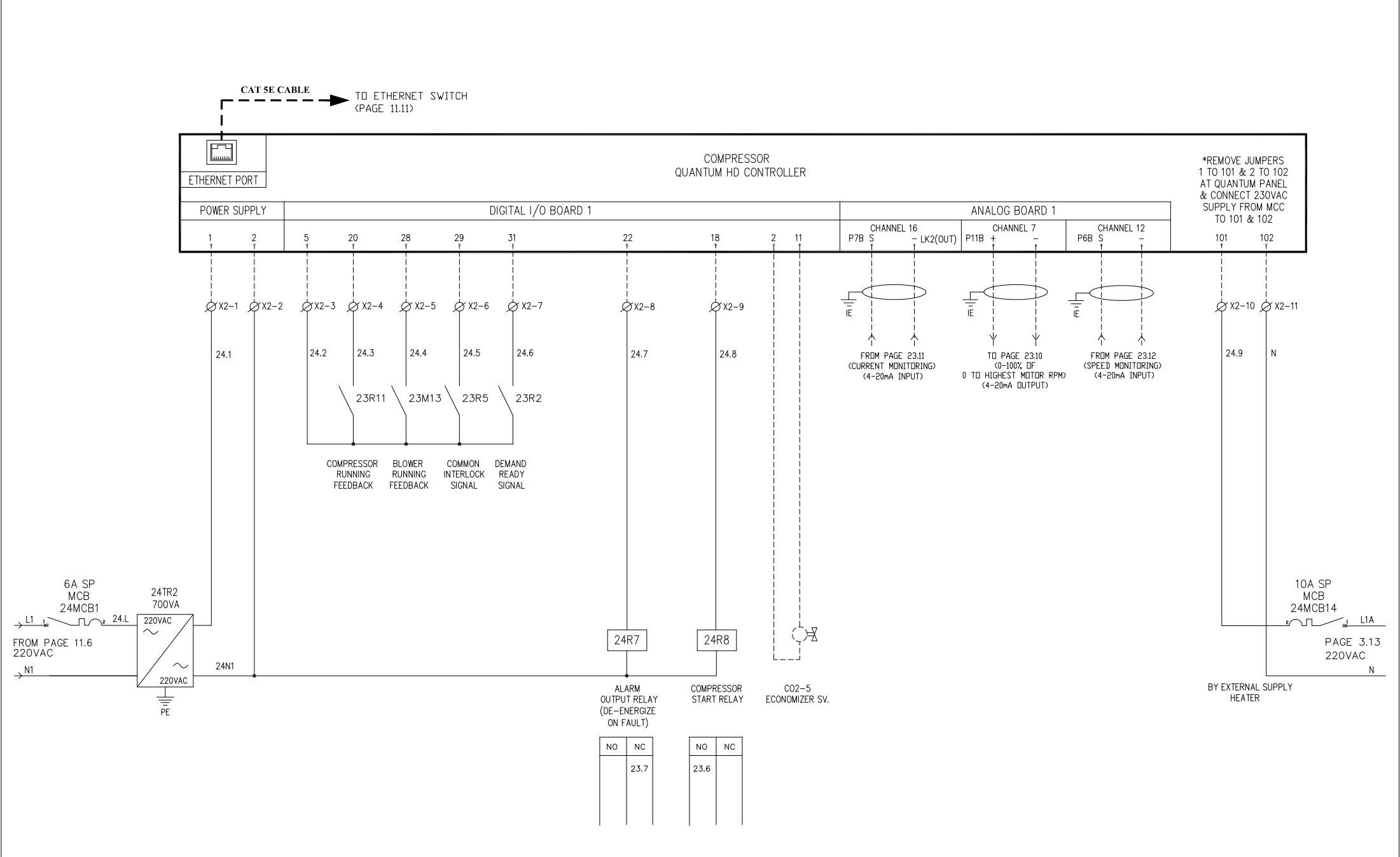


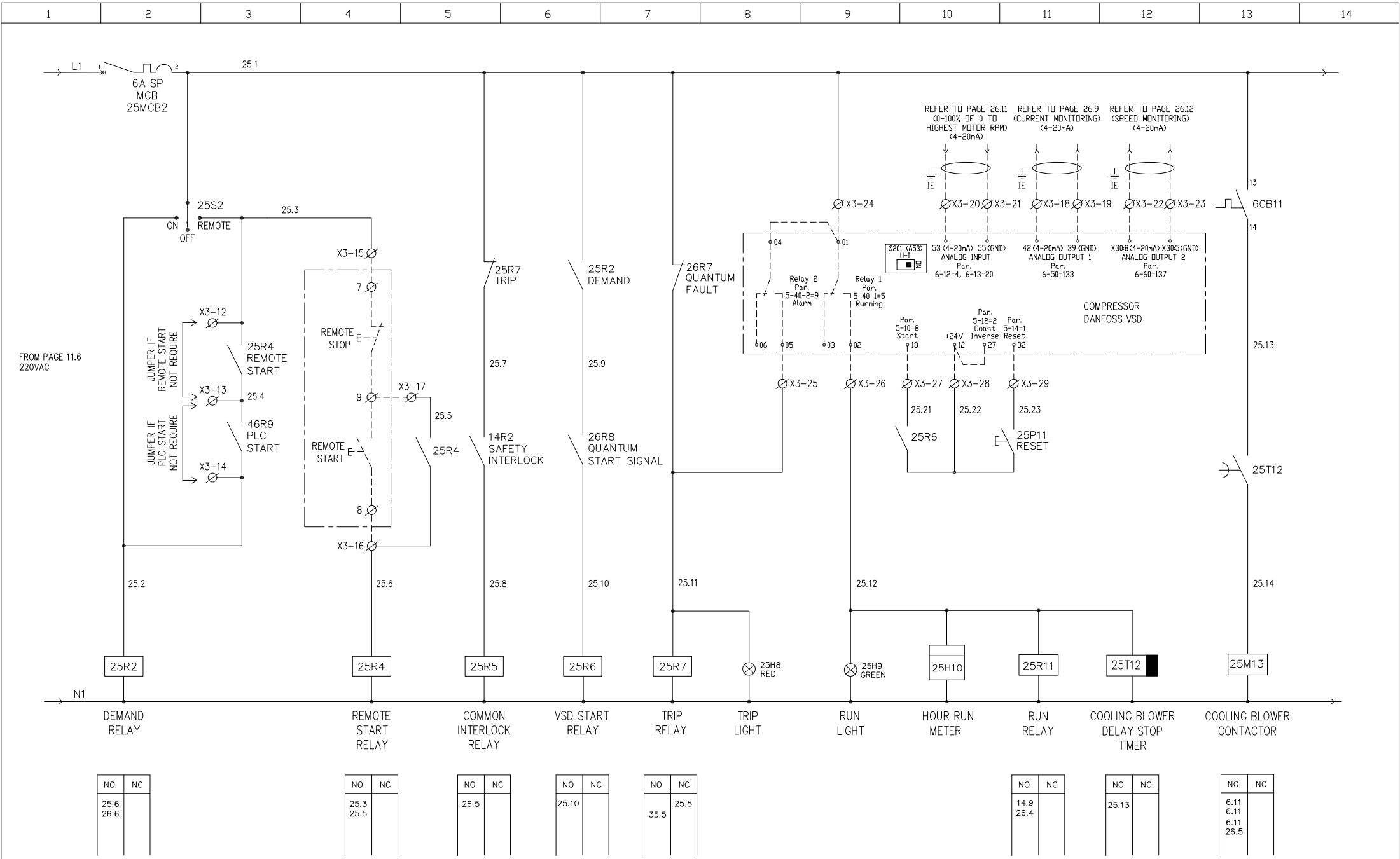


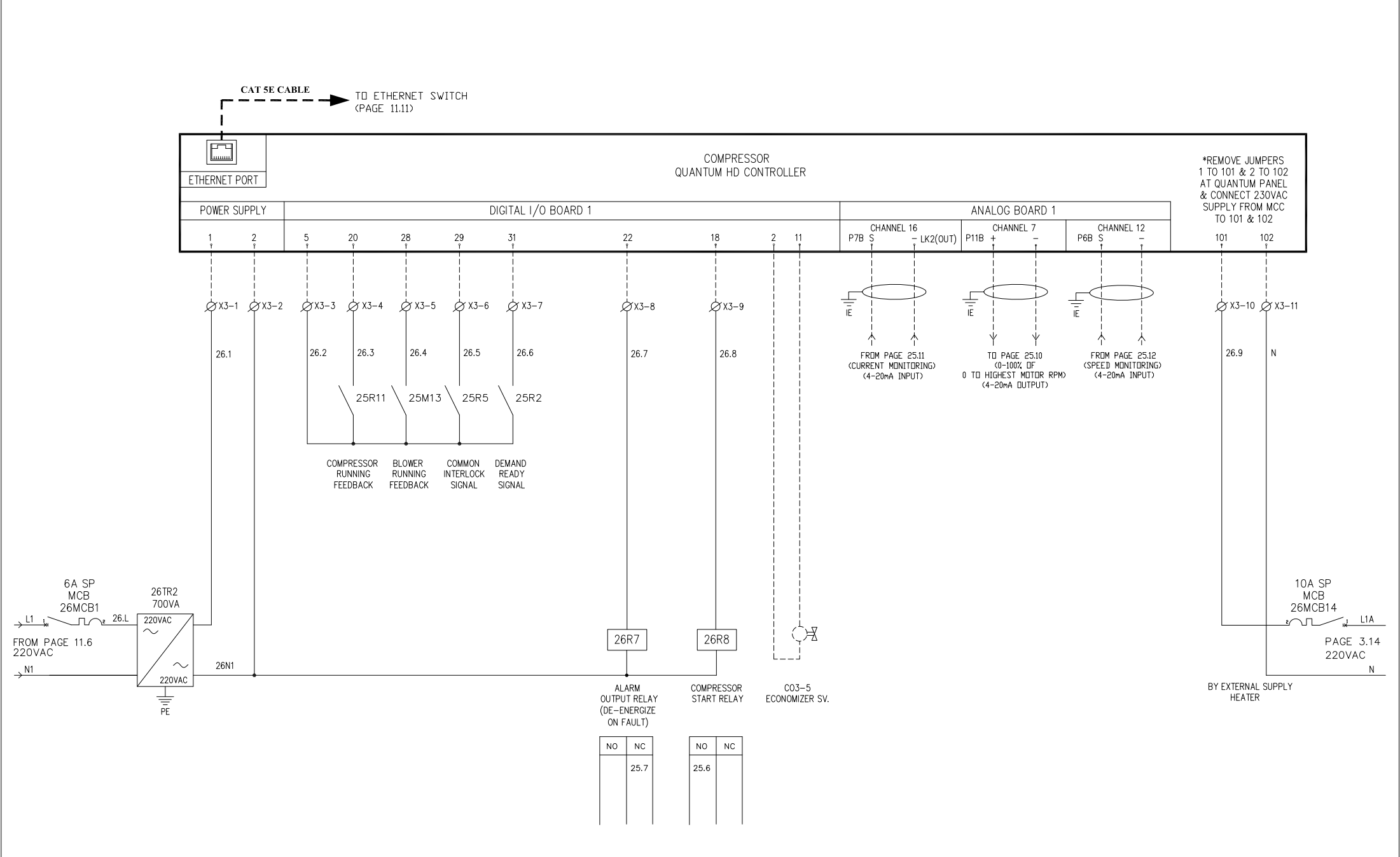
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 Johnson Controls (S) Pte. Ltd. 6 Changi Business Park Avenue 1 #06-21/22 UE Bizhub East (North Tower), Singapore 486017 Telephone: (65) 6748 0202 Telefax: (65) 6284 3473 Company Registration No: 196800276K	CLIENT DESMET BALLESTRA (MNA1 PROJECT)	TITLE BLANK	DRN. WENG	SCALE N.T.S.	DRAWING NO. 2018\JCI\DESMET-MNA1-01		PAGE NO. 15-20
			CHD. JOE	DATE APR 2018			
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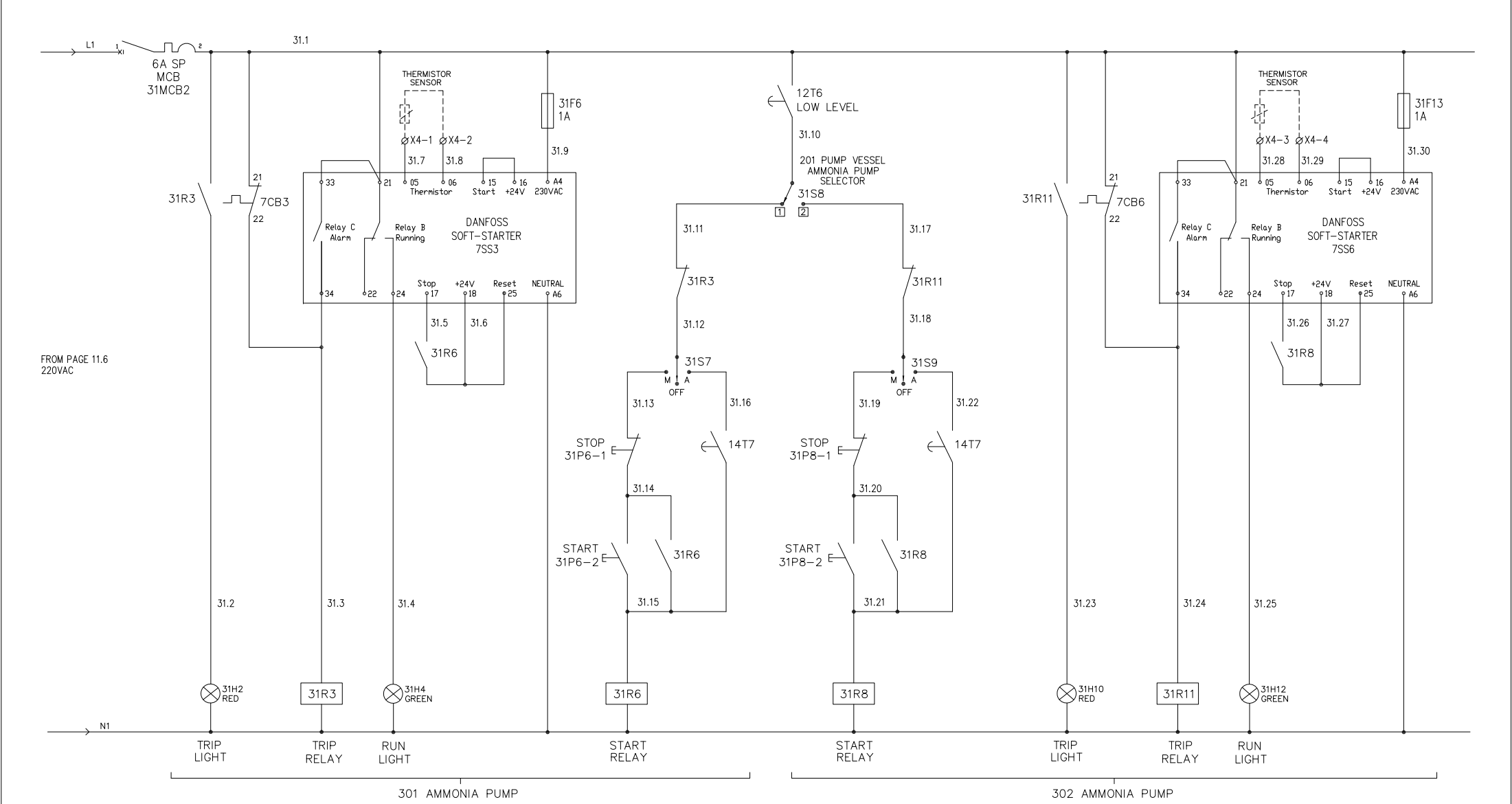




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 Johnson Controls (S) Pte. Ltd. 6 Changi Business Park Avenue 1 #06-21/22 UE Bizhub East (North Tower), Singapore 486017 Telephone: (65) 6748 0202 Telefax: (65) 6284 3473 Company Registration No: 196800276K	CLIENT DESMET BALLESTRA (MNA1 PROJECT)	TITLE BLANK	DRN. WENG	SCALE N.T.S.	DRAWING NO. 2018\JCI\DESMET-MNA1-01		PAGE NO. 27-30
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			JOB No. SG02/18-0008	REV.1 13-05-18			TOTAL PAGE 80

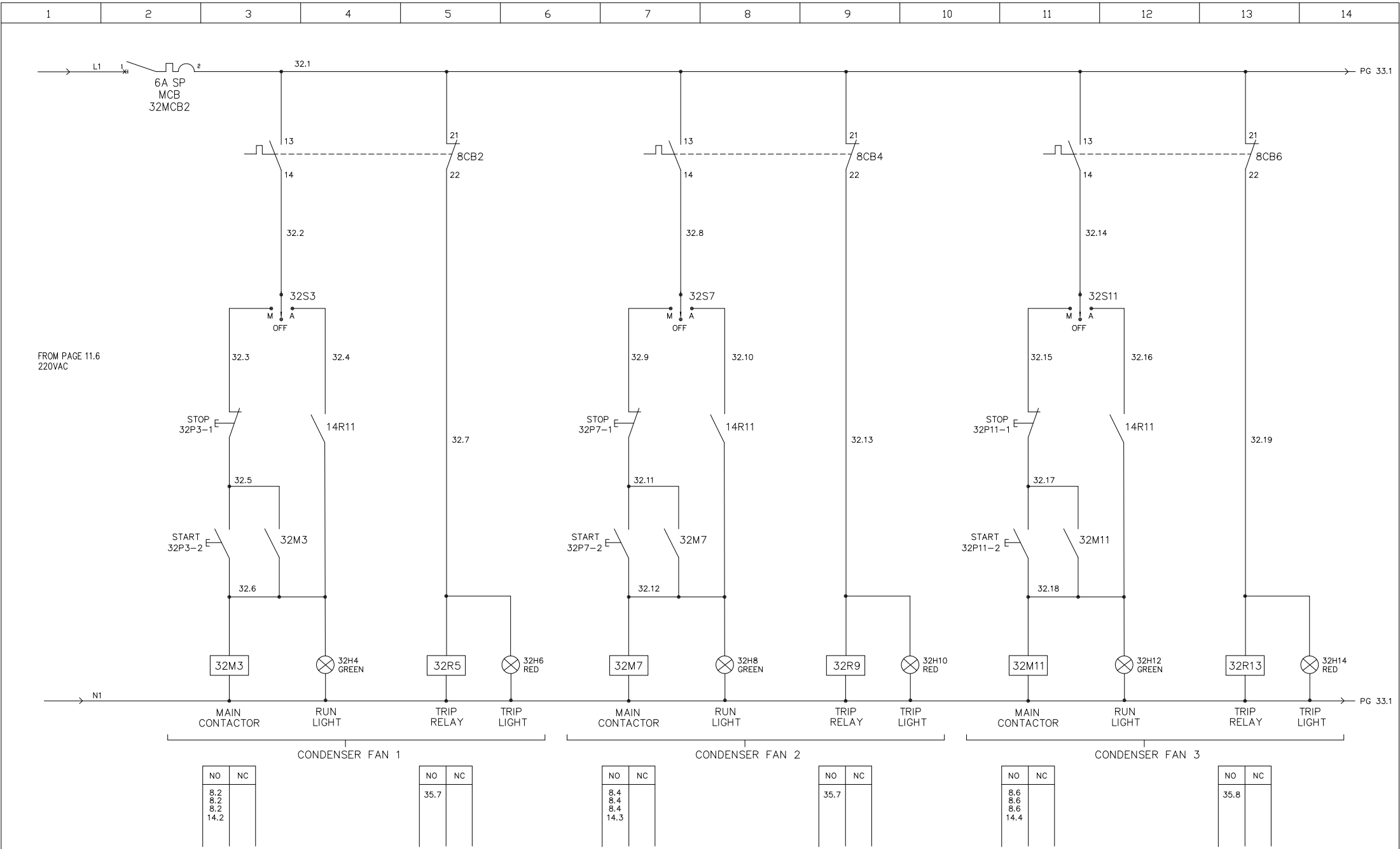


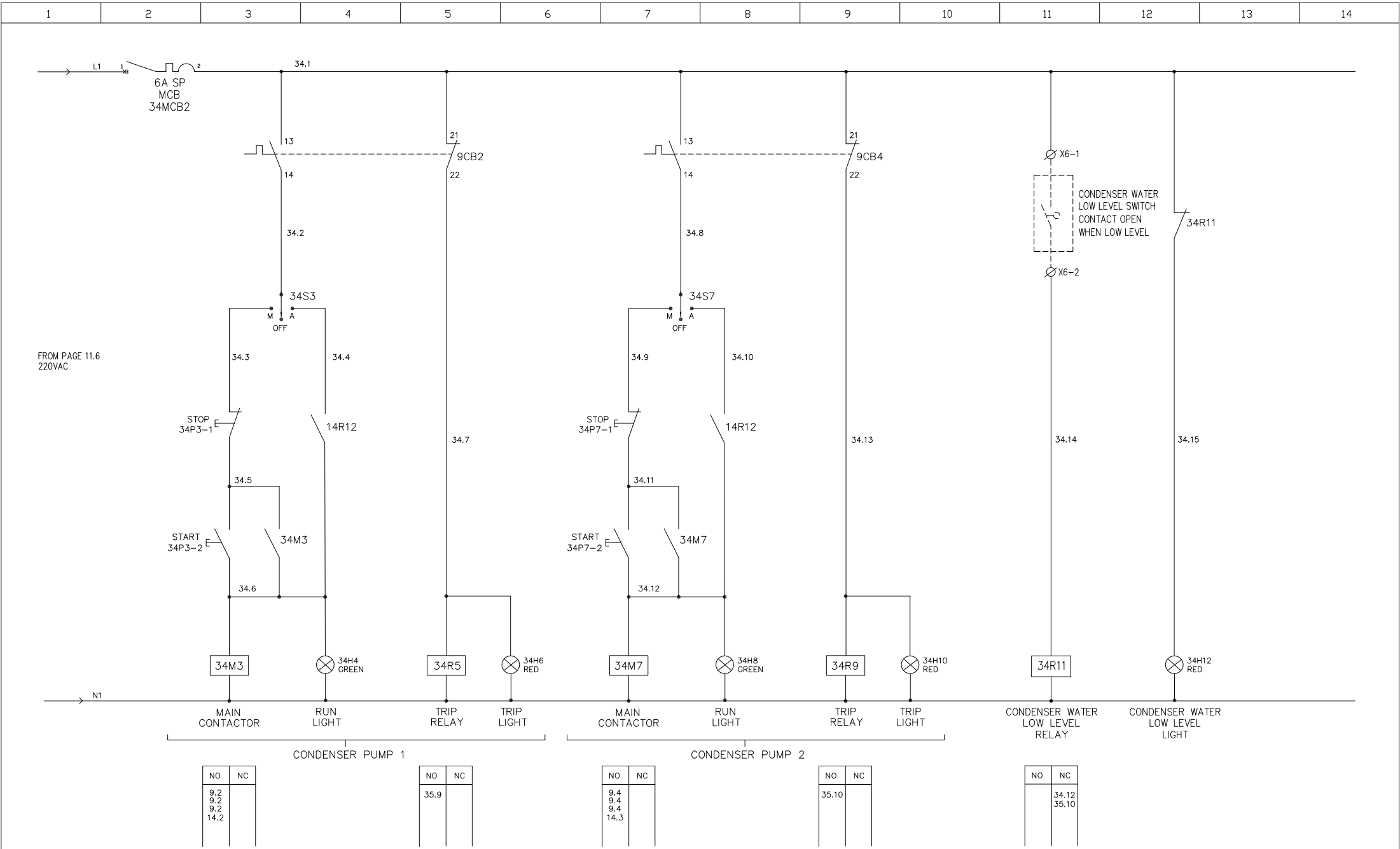
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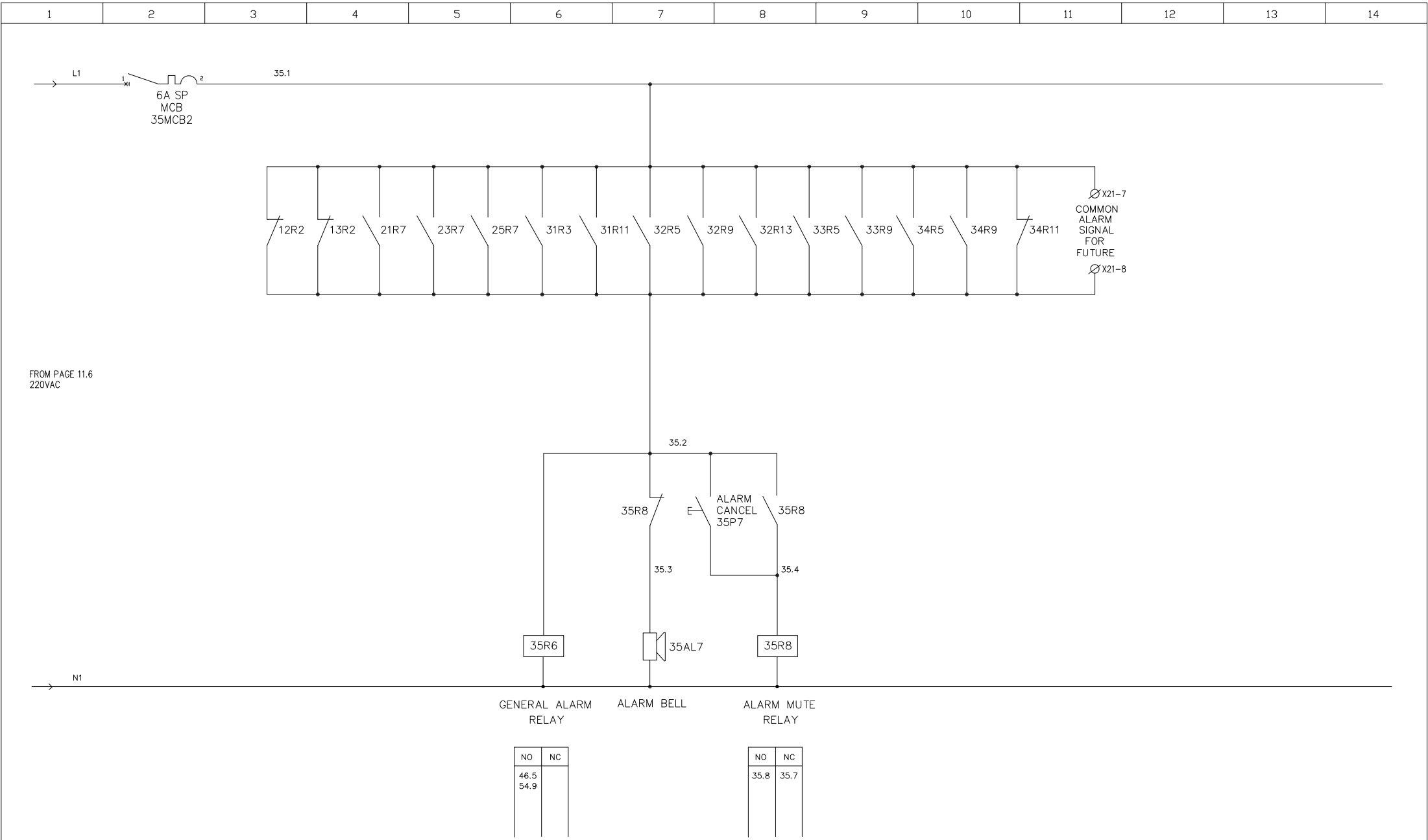
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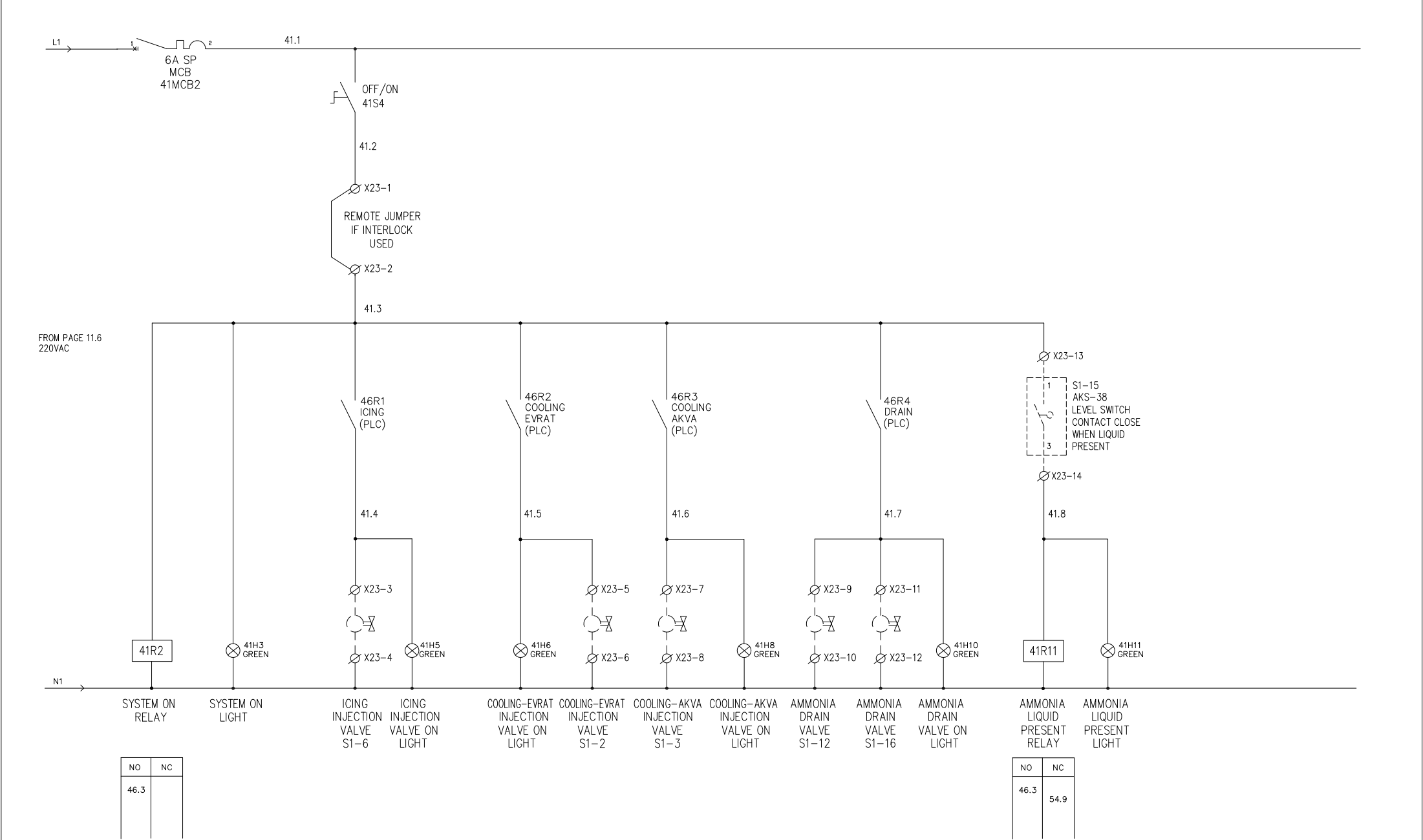


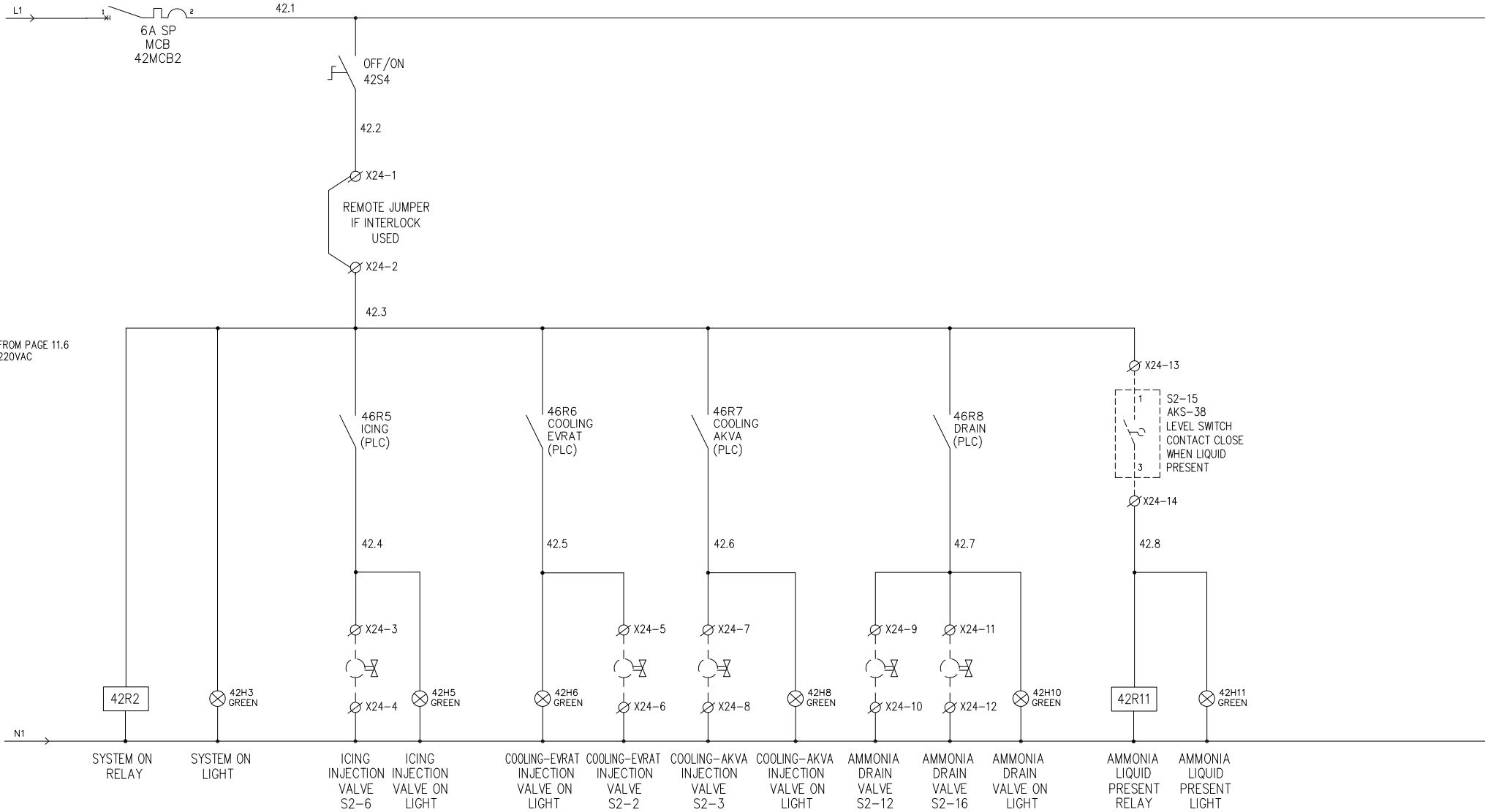
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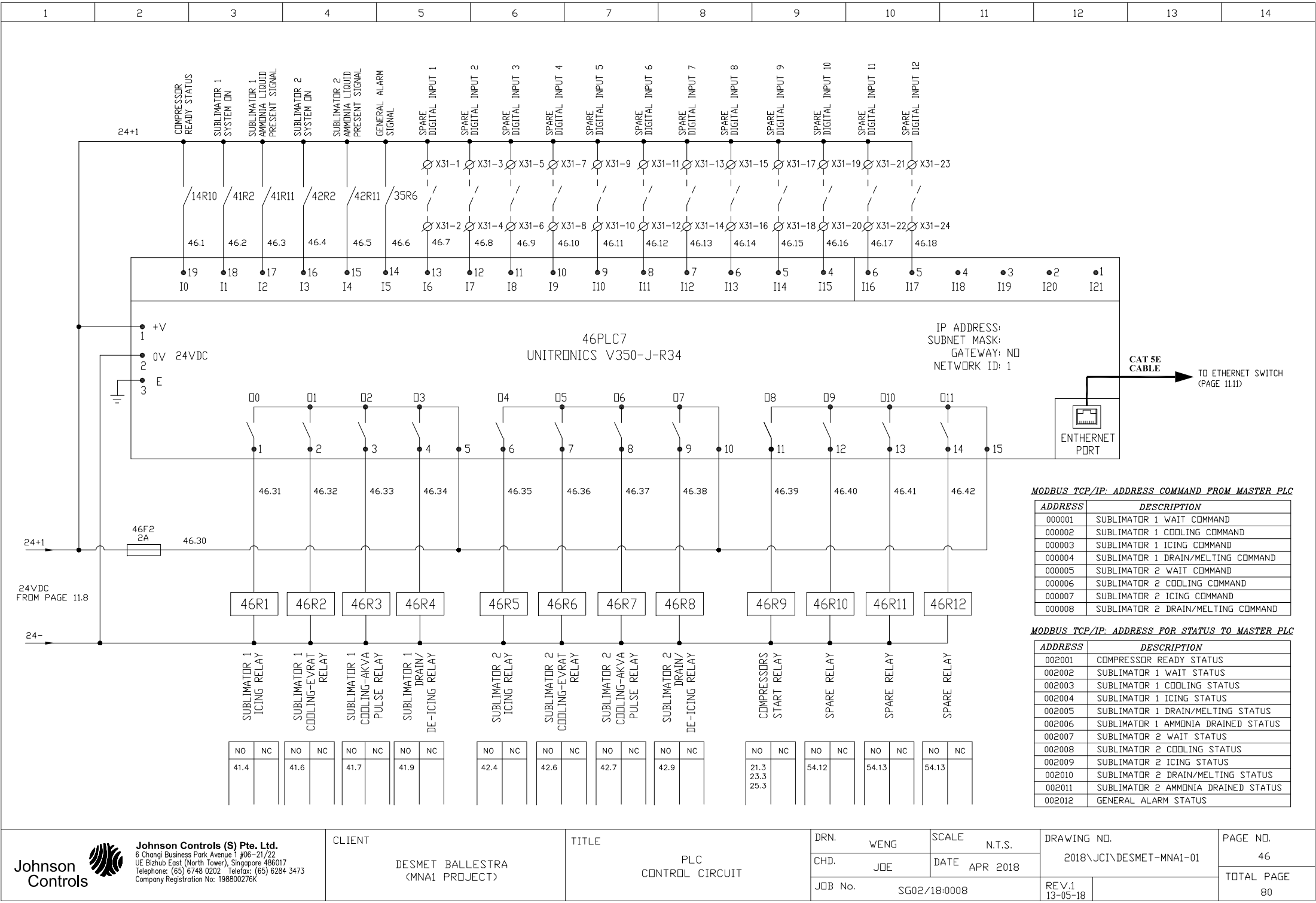
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			CHD. JOE	DATE APR 2018			TOTAL PAGE
			JOB No. SG02/18-0008		REV.1 13-05-18		80



Johnson Controls

Johnson Controls (S) Pte. Ltd.

6 Changi Business Park Avenue 1 #06-21/22

UE Bizhub East (North Tower), Singapore 486017

Telephone: (65) 6748 0202 Telefax: (65) 6284 3473

Company Registration No: 198800276K

CLIENT

DESMET BALLESTRA

(MNA1 PROJECT)

TITLE

PLC

CONTROL CIRCUIT

DRN.

WENG

SCALE

N.T.S.

CHD.

JOE

DATE

APR 2018

JOB No.

SG02/18-0008

DRAWING NO.

2018\JCI\DESMET-MNA1-01

REV.1

13-05-18

PAGE NO.

46

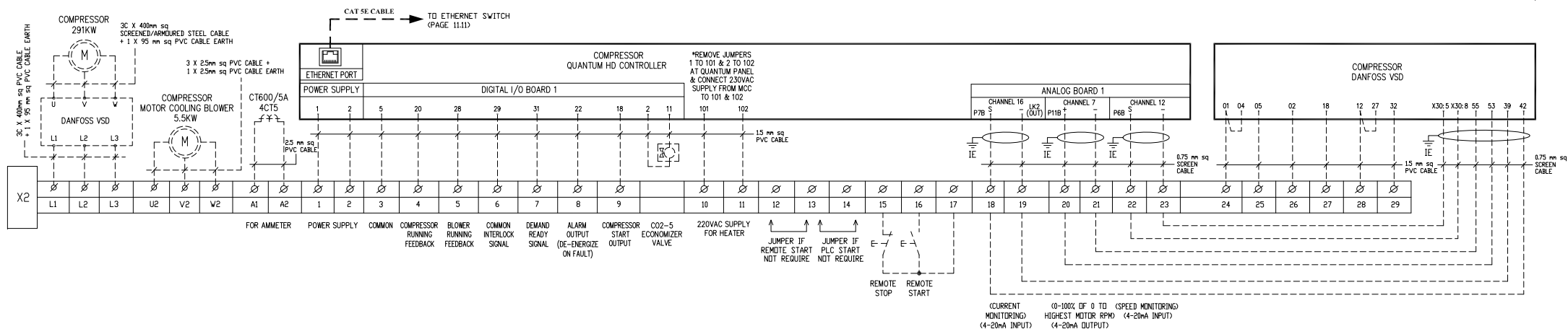
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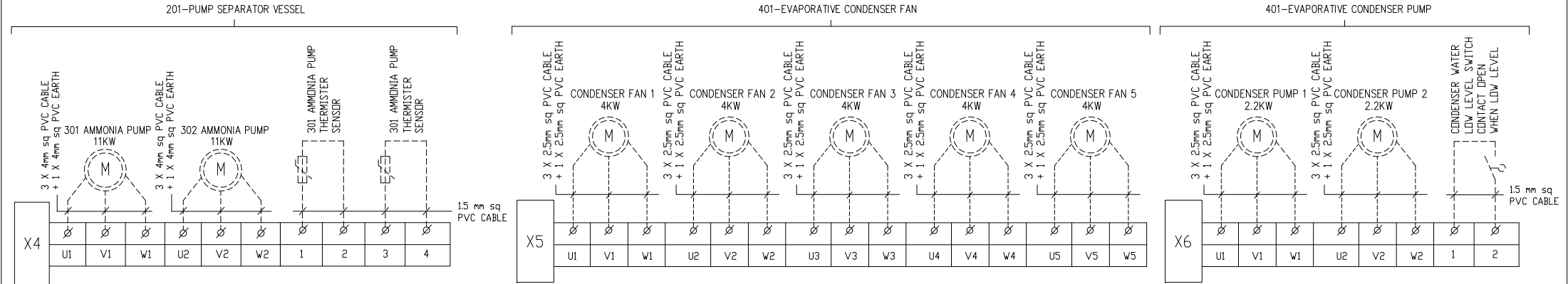
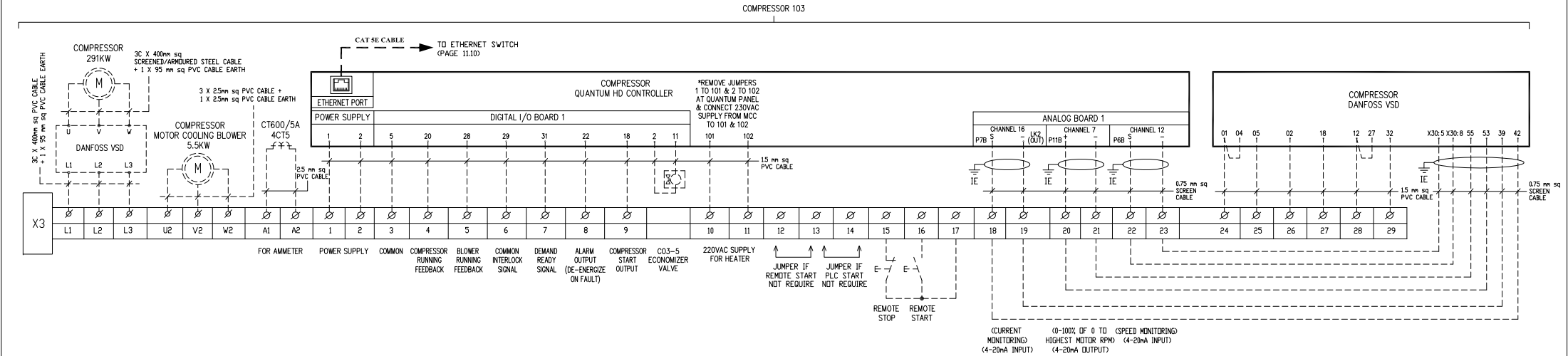
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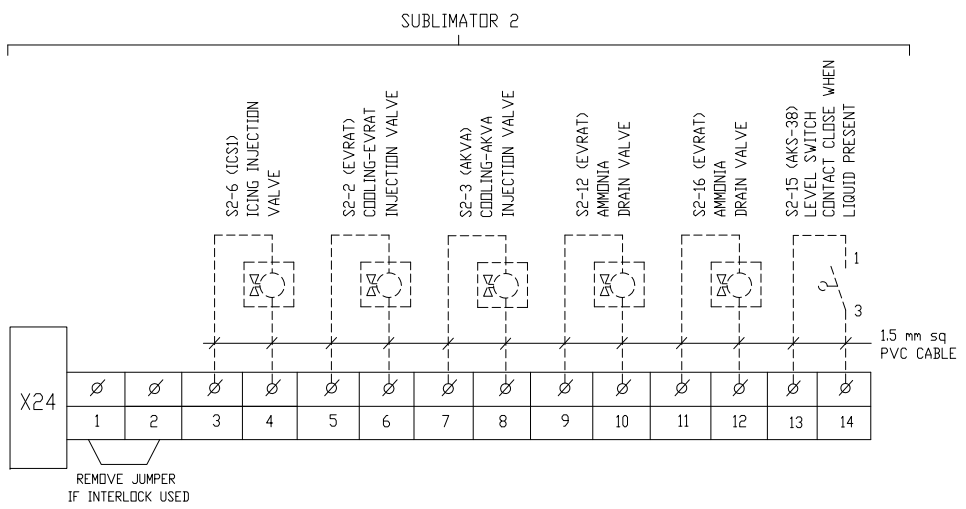
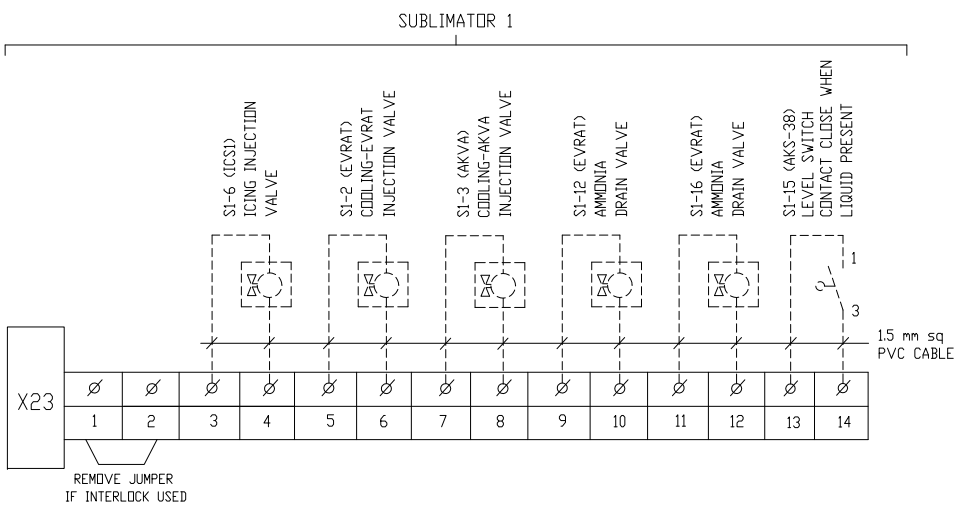
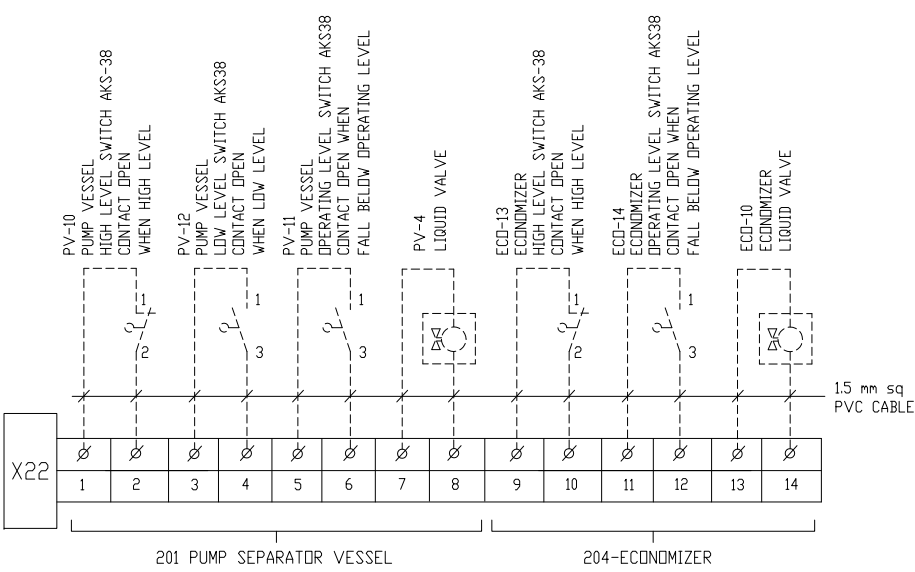
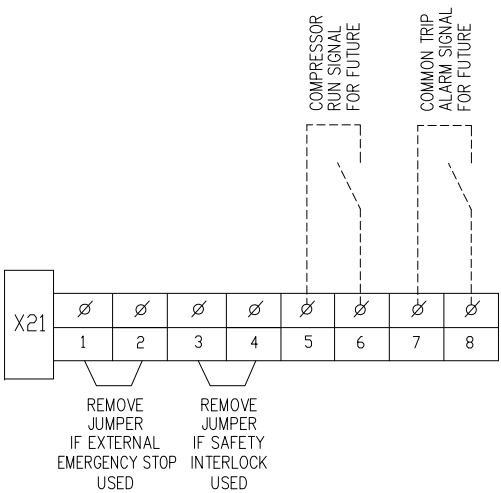
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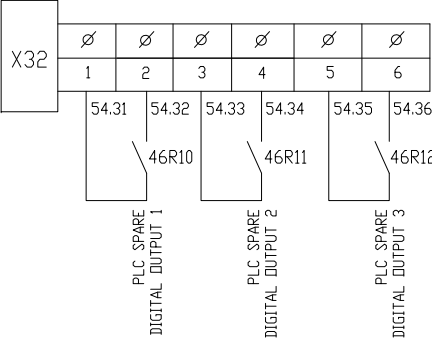
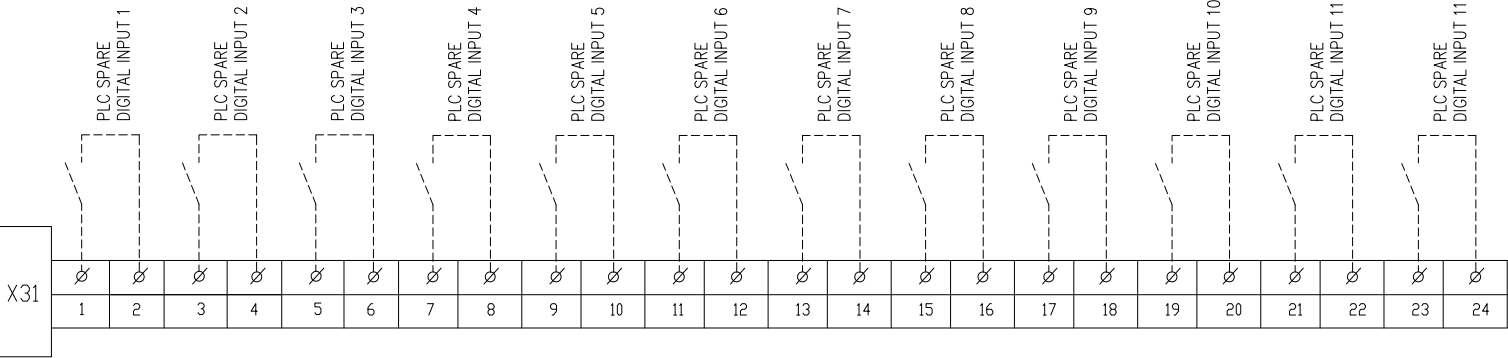
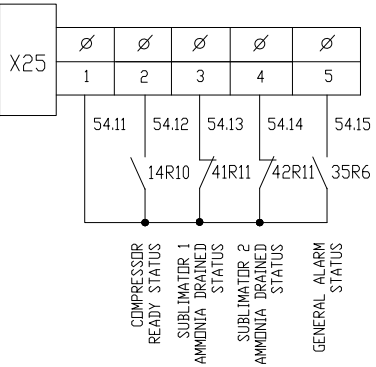
 Johnson Controls (S) Pte. Ltd. 6 Changi Business Park Avenue 1 #06-21/22 UE Bizhub East (North Tower), Singapore 486017 Telephone: (65) 6748 0202 Telefax: (65) 6284 3473 Company Registration No: 196800276K	CLIENT DESMET BALLESTRA (MNA1 PROJECT)	TITLE BLANK	DRN. WENG	SCALE N.T.S.	DRAWING NO. 2018\JCI\DESMET-MNA1-01		PAGE NO. 47-50
			CHD. JOE	DATE APR 2018			
			JOB No. SG02/18-0008	REV.1 13-05-18			TOTAL PAGE 80







(FOR HARDWIRE OPTION ONLY)
VOLTAGE FREE CONTACT INTERLOCK SIGNAL



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 Johnson Controls (S) Pte. Ltd. 6 Changi Business Park Avenue 1 #06-21/22 UE Bizhub East (North Tower), Singapore 486017 Telephone: (65) 6748 0202 Telefax: (65) 6284 3473 Company Registration No: 196800276K	CLIENT DESMET BALLESTRA (MNA1 PROJECT)	TITLE BLANK	DRN. WENG	SCALE N.T.S.	DRAWING NO. 2018\JCI\DESMET-MNA1-01		PAGE NO. 55-60
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			JOB No. SG02/18-0008	REV.1 13-05-18			TOTAL PAGE 80

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COMPRESSOR 101, 102, 103
(SETTING LIST FOR DANFOSS INVERTER)

(MOTOR DATA)

Power	291KW
Ampere	546A
Speed	3550 RPM
Frequency	50HZ
Voltage	380V

(STANDARD SETTINGS TO CHANGE)

Para./No.	Description	Factory Setting	Setting	REMARK
0-02	MOTOR SPEED UNIT	HZ	RPM (0)	
0-20	DISPLAY LINE 1:1 SMALL	REFERENCE (%)	MOTOR CURRENT (1614)	
0-21	DISPLAY LINE 1:2 SMALL	MOTOR CURRENT (A)	KWH COUNTER (1502)	
0-22	DISPLAY LINE 1:3 SMALL	POWER (KW)	RUNNING HOURS (1501)	
0-23	DISPLAY LINE 2 LARGE	FREQUENCY (HZ)	SPEED (RPM) (1617)	
0-24	DISPLAY LINE 3 LARGE	KWH COUNTER (KWH)	ANALOG OUTPUT 42 (mA) (1665)	
0-40	[HAND ON] KEY ON LCP	ENABLED	DISABLED (0)	
1-03	TORQUE CHARACTERISTICS	AUTO ENERGY OPTIM. VT	COMPRESSOR TORQUE (0)	
1-20	MOTOR POWER (KW)	315KW	315KW	
1-22	MOTOR VOLTAGE	400V	380V	
1-23	MOTOR FREQUENCY	50HZ	50HZ	
1-24	MOTOR CURRENT	580A	546A	
1-25	MOTOR NOMINAL SPEED	1495RPM	3550RPM	
1-77	COMPRESSOR START MAX SPEED	600RPM	0RPM	
1-90	MOTOR THERMAL PROTECTION	ETR TRIP 1	NO PROTECTION (0)	
3-02	MINIMUM REFERENCE	0RPM	0RPM	
3-03	MAXIMUM REFERENCE	3000RPM	3600RPM	
3-41	RAMP 1 RAMP UP TIME	90SEC	35SEC	
3-42	RAMP 1 RAMP DOWN TIME	90SEC	1SEC	
3-82	START RAMPING UP TIME	9SEC	20SEC	
4-13	MOTOR SPEED HIGH LIMIT (RPM)	1500RPM	3600RPM	
4-51	WARNING CURRENT HIGH	600A	546A	
4-53	WARNING SPEED HIGH	1500RPM	800RPM	
5-10	TERMINAL 18 DIGITAL INPUT	START	START (8)	
5-11	TERMINAL 19 DIGITAL INPUT	REVERSING	START REVERSING (11)	
5-12	TERMINAL 27 DIGITAL INPUT	COAST INVERSE	COAST INVERSE (2)	
5-14	TERMINAL 32 DIGITAL INPUT	NO OPERATION	RESET (1)	
5-19	TERMINAL 37 SAFE STOP	SAFE STOP ALARM	SAFE STOP ALARM	
5-40-1	RELAY 1 FUNCTION	CONTROL READY	RUNNING	
5-40-2	RELAY 2 FUNCTION	DRIVE READY	ALARM	
6-12	TERMINAL 53 LOW CURRENT	4mA	4mA	
6-13	TERMINAL 53 HIGH CURRENT	20mA	20mA	
6-14	TERMINAL 53 LOW REF./FEEDBACK VALUE	0	1000	
6-15	TERMINAL 53 HIGH REF./FEEDBACK VALUE	1500	3600	
6-50	TERMINAL 42 OUTPUT	OUTPUT FREQ.	MOTOR CURRENT 4-20mA (133)	
6-52	TERMINAL 42 OUTPUT MAX SCALE	100%	100%	
6-60	TERMINAL X30/8 OUTPUT	NO OPERATION	SPEED 4-20mA (137)	

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LEGEND	COMPONENT REF.	DESCRIPTION	MAKER	MODEL
	3EF6	EARTH FAULT RELAY	MH	EF18
	3CT6	PROTECTION C.T.	MH	1600/5A 5P10 15VA
	3CT11	POWER METER C.T.	MH	1600/5A CL1 5VA
	4CT5, 5CT5, 6CT5	COMPRESSOR AMMETER C.T.	MH	600/5A CL1 5VA
	4A4, 5A4, 6A4	COMPRESSOR AMMETER	MH	600/5A
	3PM11	POWER METER	MH	MD-VAFR
	21H10, 23H10, 25H10	HOUR RUN METER	MH	230VAC 48x48S
	3F2, 3F3, 3F4, 46F2	CONTROL FUSE	HAGER	2A
	3F10, 3F11, 3F12, 11F8	CONTROL FUSE	HAGER	4A
	31F6, 31F13	CONTROL FUSE	HAGER	1A
	11ST6	VOLTAGE STABILIZER	MATSUNAGA STAVOL	SVC-2000NA-BS
	11TF8	24VDC POWER SUPPLY	MEANWELL	LRS-100-24
	22TR2, 24TR2, 26TR2	STEPDOWN TRANSFORMER	ETA	220VAC/220VAC/700VA
	35AL7	ALARM BELL	CAMSCO	230V
	11FAN2	PANEL EXHAUST FAN	MAXAIR	BT220
	12R10, 13R6, 14R10, 21R5, 21R6, 22R7, 22R8, 23R5, 23R6, 24R7, 24R8, 25R5, 25R6, 26R7, 26R8, 32R5, 32R9, 32R13, 33R5, 33R9, 34R5, 34R9, 35R6, 35R8, 41R2, 42R2	230VAC CONTROL RELAY	IDEC IZUMI	RU2S-NF-A220 (230VAC)
	12R2, 12R5, 13R2, 14R2, 14R11, 14R12, 21R2, 21R4, 21R7, 21R11, 23R2, 23R4, 23R7, 23R11, 25R2, 25R4, 25R7, 25R11, 31R3, 31R6, 31R8, 31R11, 34R11, 41R11, 42R11,	230VAC CONTROL RELAY	IDEC IZUMI	RU4S-NF-A220 (230VAC)
	46R1, 46R2, 46R3, 46R4, 46R5, 46R6, 46R7, 46R8, 46R10, 46R11, 46R12	24VDC CONTROL RELAY	IDEC IZUMI	RU2S-NF-D24 (24VDC)
	46R9	24VDC CONTROL RELAY	IDEC IZUMI	RU4S-NF-D24 (24VDC)
	12T6, 12T9, 13T5, 14T3, 14T7, 14T8,	230VAC ON DELAY TIMER	IDEC IZUMI	GE1A-C30HA (230VAC)
	14T9, 21T12, 23T12, 25T12	230VAC OFF DELAY TIMER	IDEC IZUMI	GT3F-2AF20 (230VAC)
	ALL LIGHT	PILOT LIGHT	SCHNEIDER	XB7-EV0.MP
	11ES4	EMERGENCY STOPPUSH BUTTON	IDEC IZUMI	YW1B-V4E01R
	12P3/12H4, 13P3/13H4	PUSH BUTTON WITH PILOT LIGHT	SCHNEIDER	XB5AW34M5-RED
	11SD9	SWITCH SOCKET OUTLET	MK	13A SS0

 Johnson Controls (S) Pte. Ltd. 6 Changi Business Park Avenue 1 #06-21/22 UE Bizhub East (North Tower), Singapore 486017 Telephone: (65) 6748 0202 Telefax: (65) 6284 3473 Company Registration No: 198800276K	CLIENT DESMET BALLESTRA (MNA1 PROJECT)	TITLE MATERIAL LIST	DRN. WENG	SCALE N.T.S.	DRAWING NO. 2018\JCI\DESMET-MNA1-01	PAGE NO. 63
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			JOB No. SG02/18-0008		REV.1 13-05-18	TOTAL PAGE 80

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LEGEND	COMPONENT REF.	DESCRIPTION	MAKER	MODEL
	21P11, 23P11, 25P11, 31P6-2, 31P8-2, 32P3-2, 32P7-2, 32P11-2, 33P3-2, 33P7-2, 34P3-2, 34P7-2, 35P7	NORMALLY OPEN PUSH BUTTON	IDEC IZUMI	YW-E10
	31P6-1, 31P8-1, 32P3-1, 32P7-1, 32P11-1, 33P3-1, 33P7-1, 34P3-1, 34P7-1	NORMALLY CLOSE PUSH BUTTON	IDEC IZUMI	YW-E01
	31S8, 41S4, 42S4	ON / OFF SELECTOR SWITCH	IDEC IZUMI	YW-1S 2E10
	21S2, 23S2, 25S2, 31S7, 31S9, 32S3, 32S7, 32S11, 33S3, 33S7, 34S3, 34S7	SELECTOR SW. (3 POSITION)	IDEC IZUMI	YW-1S 3E20
	3MCCB5	MAIN MCCB	SCHNEIDER	NS1600N 3POLES 1600A C/W SHT
	4MCCB5, 5MCCB5, 6MCCB5	MCCB	SCHNEIDER	NSX630N 3 POLES 630A
	11MCB8	CONTROL MCB	SCHNEIDER	C60NC 6A 2P
	11MCB6	CONTROL MCB	SCHNEIDER	C60ND 10A 2P
	11MCB6-1	CONTROL MCB	SCHNEIDER	C60NC 10A 2P
	11MCB2, 11MCB4, 11MCB9, 12MCB2, 21MCB2, 22MCB1, 23MCB2, 24MCB1, 25MCB2, 26MCB1, 31MCB2, 32MCB2, 34MCB2, 35MCB2, 41MCB2, 42MCB2	CONTROL MCB	SCHNEIDER	C60NC 6A 1P
	22CB14, 24MCB14, 26MCB14	CONTROL MCB	SCHNEIDER	C60NC 10A 1P
	11RCCB9	RESIDUAL CURRENT CIRCUIT BREAKER	SCHNEIDER	40A DP 30mA
	11M5, 34M3, 34M7	CONTACTOR	SCHNEIDER	LC1-D09
	32M3, 32M7, 32M11, 33M3, 33M7	CONTACTOR	SCHNEIDER	LC1-D12
	21M13, 23M13, 25M13	CONTACTOR	SCHNEIDER	LC1-D18
	9CB2, 9CB4	MOTOR CIRCUIT BREAKER	SCHNEIDER	GV2ME10 + GVAE11
	8CB2, 8CB4, 8CB6, 8CB8, 8CB10	MOTOR CIRCUIT BREAKER	SCHNEIDER	GV2ME14 + GVAE11
	4CB11, 5CB11, 6CB11	MOTOR CIRCUIT BREAKER	SCHNEIDER	GV2ME16 + GVAE11
	7CB3, 7CB6	MOTOR CIRCUIT BREAKER	SCHNEIDER	GV2ME22 + GVAE11
	11MOV7, 11MOV7-1	VARIATOR	SIEMENS	K250
	11ETS11	ETHERNET SWITCH	TP LINK	TL-SF1005D
	5SS3, 5SS6	SOFT-STARTER	DANFOSS	MCD5-0037B
	46PLC7	PLC CONTROLLER	UNITRONICS	V350-J-R34

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			CHD. JOE	DATE APR 2018			
			JOB No. SG02/18-0008	REV.1 13-05-18			TOTAL PAGE 80